

MYOGLOBIN & HEMOGLOBIN

Structure • Oxygen Binding • CO₂ Transport • Allostery • Clinical Biochemistry

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Learning objectives

- ❖ Explain the four levels of protein structure and relate them to globin function.
- ❖ Describe the molecular architecture of heme, myoglobin and adult hemoglobin (HbA).
- ❖ Explain oxygen binding, cooperativity, the $T \leftrightarrow R$ transition and the oxygen dissociation curve.
- ❖ Explain the Bohr effect, Haldane effect, 2,3-BPG, temperature and chloride effects.
- ❖ Describe major hemoglobin derivatives and clinically important hemoglobinopathies.
- ❖ Explain pulmonary surfactant composition, molecular structure, function and clinical relevance.

❖ *Core idea: structure determines ligand binding, and ligand binding changes structure.*

Lecture map

Unit	Key concepts
1. Protein architecture	Primary → secondary → tertiary → quaternary structure
2. Heme & myoglobin	Porphyrin, Fe ²⁺ , proximal/distal histidine, O ₂ storage
3. Hemoglobin	$\alpha_2\beta_2$ tetramer, heme sites, allostery and cooperativity
4. Oxygen transport	Dissociation curve, P50, tissue unloading
5. Allosteric regulation	2,3-BPG, Bohr effect, temperature, chloride
6. CO ₂ transport	Bicarbonate, carbaminohemoglobin, Haldane effect
7. Clinical biochemistry	COHb, metHb, HbA1c, HbS/HbC, thalassemia
8. Lung surfactant	DPPC, SP-A/B/C/D, alveolar stability, RDS

Part I — Protein structure

- ❖ Why the three-dimensional architecture of globins matters
 - It creates the heme-binding pocket:
 - The heme group sits inside a hydrophobic pocket.
 - The proximal histidine (His F8) directly coordinates the Fe^{2+} atom.
 - The distal histidine (His E7) helps control the environment around the oxygen-binding site.
 - This arrangement allows oxygen to bind reversibly rather than permanently.
 - Structure → correct heme positioning → reversible O_2 binding.
 - It controls oxygen affinity
 - The 3D arrangement of amino acids around the heme determines how easily O_2 can bind and dissociate.
 - When O_2 binds: Fe^{2+} moves slightly → proximal His moves → F-helix moves → globin conformation changes.

Why the three-dimensional architecture of globins matters Cont...

➤ It allows hemoglobin subunits to communicate:

- This is particularly important for **hemoglobin**, which consists of four globin chains: **HbA = $\alpha_2\beta_2$**
- The 3D arrangement of these four subunits creates interfaces through which one subunit can influence another.
- Therefore: **O₂ binds to one heme → structural change → neighboring subunits become more likely to bind O₂.**
- This produces **positive cooperativity** and explains the characteristic **sigmoidal oxygen dissociation curve** of hemoglobin.

➤ It creates allosteric binding sites

- The 3D architecture also creates spaces that can bind regulatory molecules.
- For example, 2,3-BPG binds within the central cavity of deoxyhemoglobin.
- Its binding stabilizes the T state and decreases O₂ affinity.
- Thus: Globin architecture → central cavity → 2,3-BPG binding → reduced O₂ affinity → enhanced tissue O₂ unloading.

Why the three-dimensional architecture of globins matters Cont...

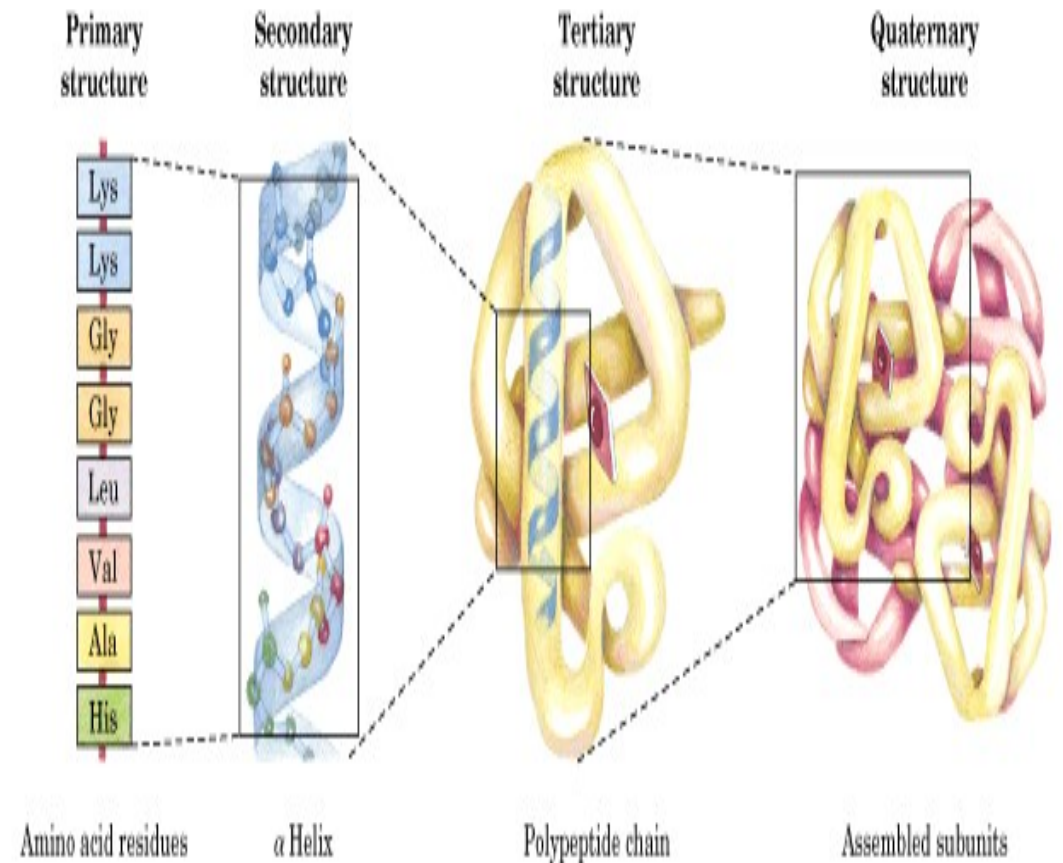
➤ Small structural changes can have major clinical effects

- A change in only one amino acid can alter the 3D properties of hemoglobin.
- For example, in **HbS**, the β -globin mutation is: **$\beta 6$ Glu $\rightarrow$ Val**
- Replacing negatively charged glutamate with hydrophobic valine creates a new hydrophobic interaction surface.
- Under deoxygenated conditions, HbS molecules can polymerize, contributing to red-cell sickling.
- So: One **amino-acid change** $\rightarrow$ **altered molecular interactions** $\rightarrow$ **altered protein behavior** $\rightarrow$ **altered cell shape** $\rightarrow$ **disease.**

❖ Primary structure determines the **folding pattern**; the 3D structure creates **functional sites**; these sites **determine ligand binding, regulation, cooperativity and ultimately physiological function.**

Levels of protein organization

Level	Structural feature	Main stabilizing forces	Example in globins
Primary	- Amino-acid sequence - Determines chemical properties and folding	Peptide bonds	α - and β -globin sequences
Secondary	α-helix, β-sheet, turns - Provides the α-helical framework	Backbone H-bonds	Globins are predominantly α -helical
Tertiary	3-D fold of one chain - Creates the heme pocket	Hydrophobic, H-bonds, ionic, van der Waals; disulfide when present	Myoglobin fold
Quaternary	- Association of multiple chains - Enables subunit communication and cooperativity	Hydrophobic, ionic, H-bonds; sometimes covalent links	HbA $\alpha_2\beta_2$

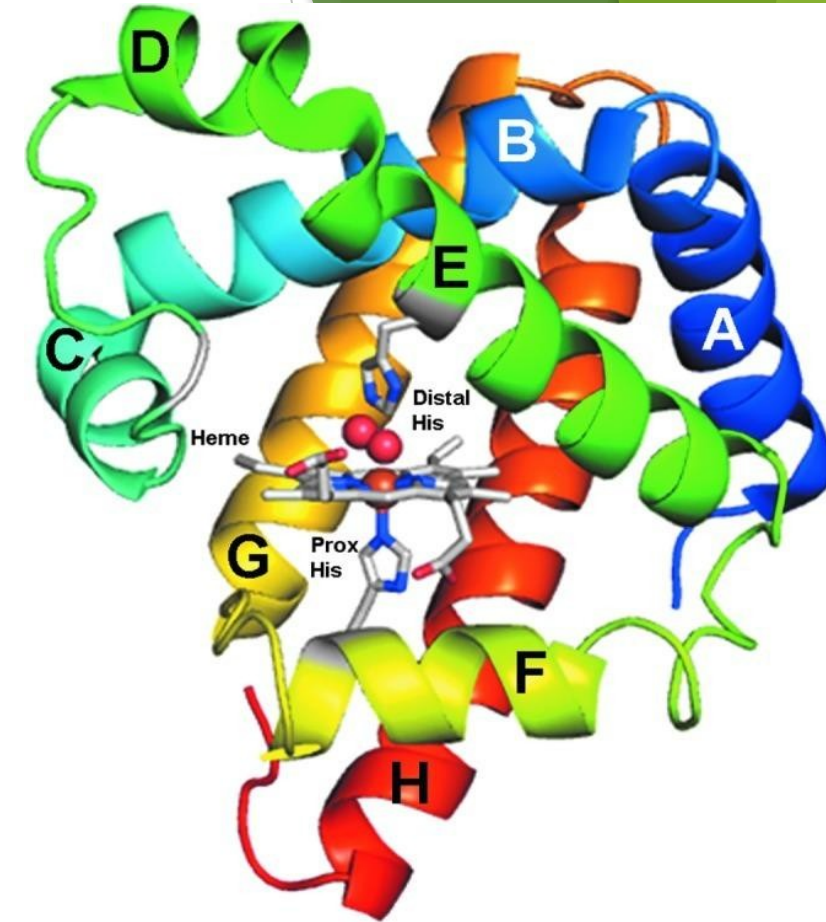


❖ Dynamic conformational changes: Allow O_2 loading/unloading and allosteric regulation

❖ A protein's sequence constrains its fold; the fold creates the heme pocket; the pocket controls ligand binding.

Secondary structure of Globins

- Globin domains are rich in α -helices rather than β -sheets.
- The helices are commonly designated A–H in the globin fold.
- Peptide backbone hydrogen bonds stabilize each α -helix.
- Nonpolar side chains pack into the interior, creating a hydrophobic core.
- Polar and charged residues are preferentially exposed to solvent, although exceptions occur where they have functional roles.
- The heme pocket is formed by the spatial arrangement of residues from several helices—not by a single continuous sequence.



❖ *Globin architecture is a classic example of how tertiary structure creates a ligand-binding pocket.*

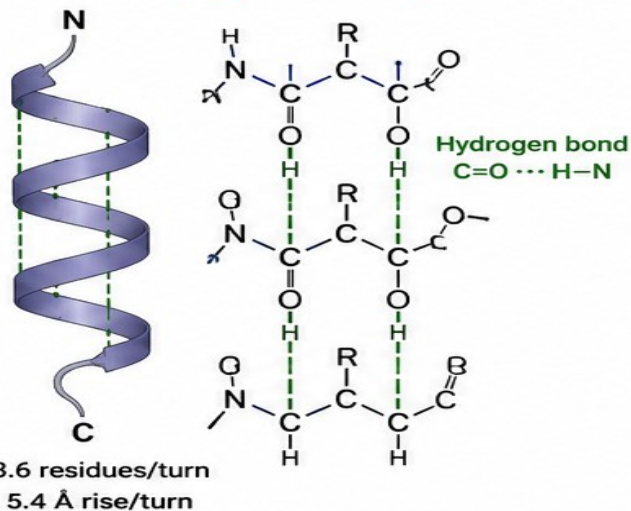
Secondary Structure of Globins

Globins are composed almost entirely of **α -helices** connected by short loops (turns).

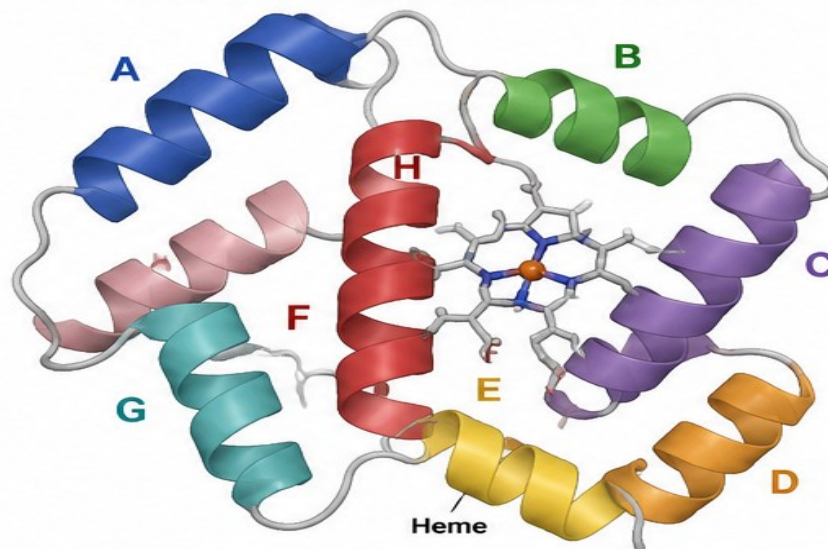
What is Secondary Structure?

- Secondary structure refers to the local, regular conformation of the polypeptide backbone stabilized by **hydrogen bonds**.
- In globins, > 75–80% of the amino acids are arranged in **α -helices**.
- These α -helices are labeled **A through H (A–H)**.

α -Helix (right-handed)



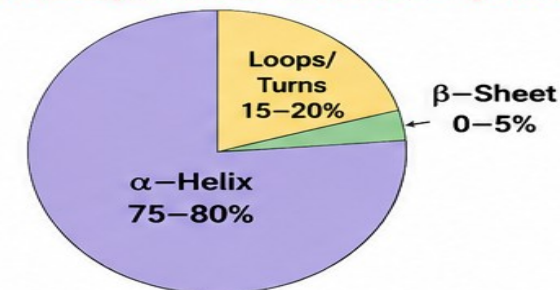
Myoglobin (example of globin fold)



Helix	Residue range (myoglobin)	Location/role
A	1 – 25	Upper left; contributes to core
B	26 – 41	Top; part of heme pocket
C	42 – 58	Right side
D	59 – 72	Lower right
E	73 – 84	Bottom
F	85 – 97	Central; contains proximal His (F8)
G	98 – 110	Lower left
H	111 – 153	Left side; forms part of heme pocket

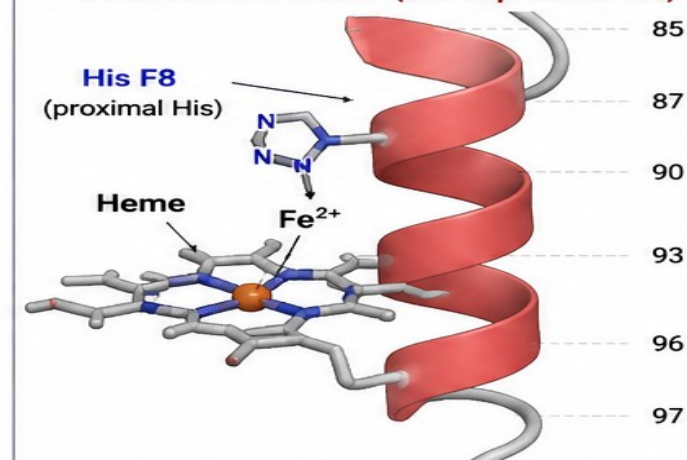
The helices (A–H) are connected by short loops (turns) that position the helices in the correct 3-D arrangement.

Secondary structure content of globins



Globins are almost entirely α -helical proteins.

Detailed view: Helix F (site of proximal His)



Helix F contains **His F8**, the proximal histidine that coordinates the Fe^{2+} of the heme.

KEY POINTS

- Globins are predominantly α -helical proteins.
- Eight α -helices (A–H) form the globin fold.
- Helices are stabilized by intrachain hydrogen bonds.
- Helices and loops create the heme pocket and functional sites.
- This secondary structure is the foundation for **tertiary and quaternary structure and function**.

Tertiary structure: The heme pocket

- The tertiary fold positions the heme deep within a largely hydrophobic pocket.
- Hydrophobic packing helps exclude bulk water and supports reversible O_2 binding.
- Proximal histidine provides the direct protein–iron coordination bond.
- Distal histidine does not directly coordinate Fe^{2+} ; it helps orient and stabilize bound O_2 and limits inappropriate ligand chemistry.
- Small conformational movements around the heme are transmitted to the globin scaffold.

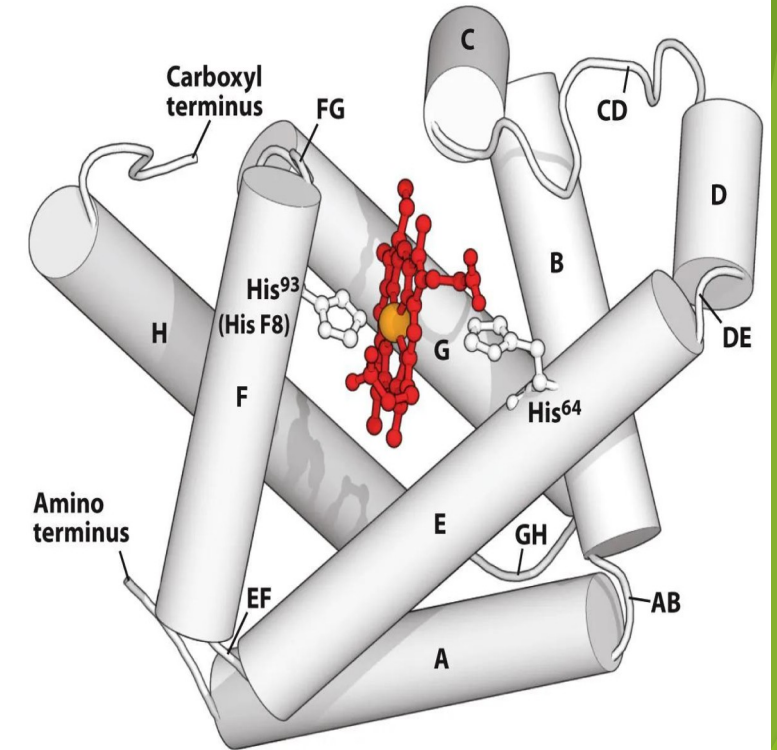


Figure 5-3
Lehninger Principles of Biochemistry, Seventh Edition
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❖ Key structural principle

- The heme is not simply 'floating' in the protein. Specific amino-acid residues create a controlled microenvironment for Fe^{2+} and O_2 .

Tertiary Structure: The Heme Pocket of Globins

The globin polypeptide folds in three dimensions to create a hydrophobic **heme pocket** in which the heme group is precisely positioned for ligand binding.

1. What is Tertiary Structure?

- Tertiary structure refers to the overall 3D conformation of a single polypeptide chain.
- It is stabilized by interactions among amino acid side chains that are far apart in the primary sequence.
- In globins, the tertiary structure forms a compact globular fold with a **heme pocket** buried inside.
- This pocket protects the heme and creates the proper environment for reversible binding of O_2 and other ligands.

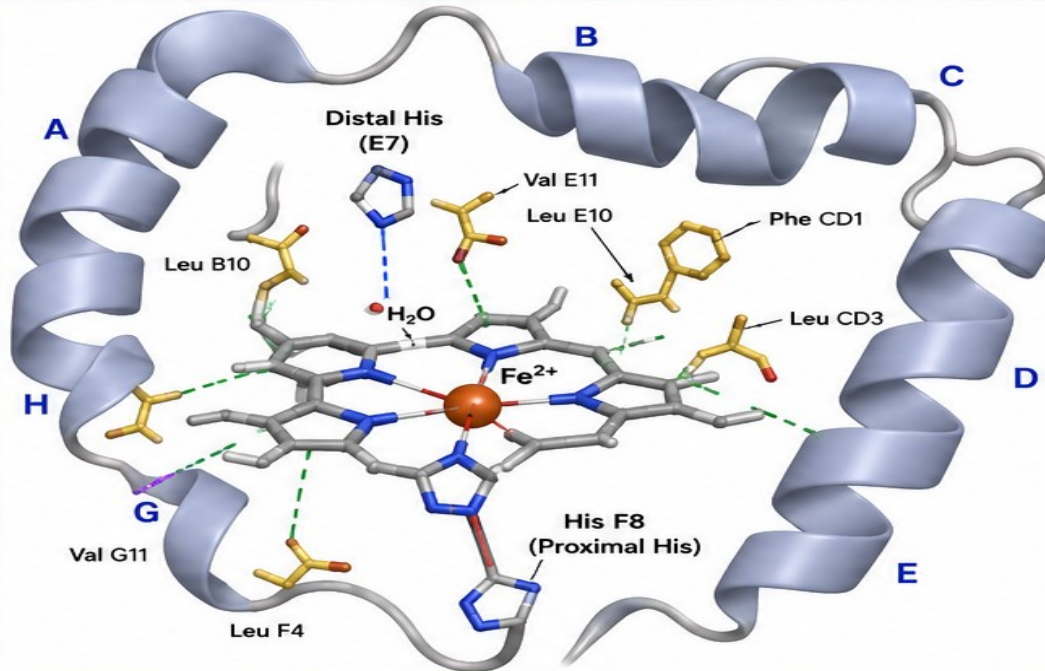
2. The Heme Pocket – Key Features

- Located between helices **E** and **F**.
- Predominantly hydrophobic interior (Val, Leu, Ile, Phe, Met).
- Shields the heme from water and prevents oxidation of Fe^{2+} .
- Provides a site for ligand binding at the sixth coordination position of Fe^{2+} .
- Small conformational changes in nearby residues control ligand access and affinity.

KEY INTERACTIONS IN THE POCKET

- Hydrogen bond
- Hydrophobic interaction
- Coordination bond
- Ionic interaction

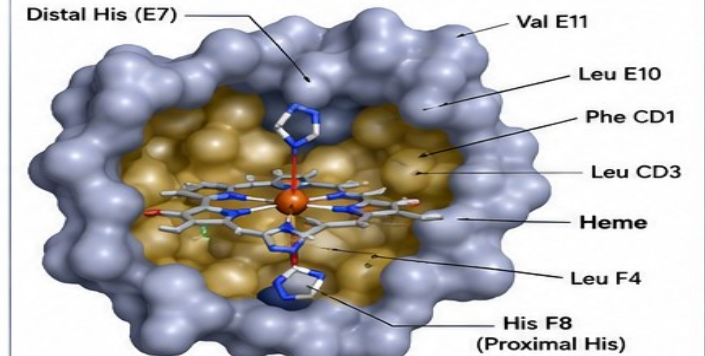
3. Myoglobin – Tertiary Structure and the Heme Pocket



- The heme is held in place by hydrophobic interactions and van der Waals forces.
- The Fe^{2+} is coordinated by 4 porphyrin nitrogens (in-plane) and by **His F8** (below the plane).
- The sixth coordination site (above the plane) binds O_2 , H_2O , CO , NO , etc.

The tertiary structure of globins precisely arranges amino acid residues to form the heme pocket and control ligand binding.

4. Cross-section of the Heme Pocket (View between Helices E and F)



Hydrophobic residues line the pocket (gold). Polar residues are excluded from the interior.

5. Functional Importance of the Heme Pocket

- Protects Fe^{2+} from oxidation.
- Provides steric and electronic environment for reversible O_2 binding.
- Distal His (E7) stabilizes bound O_2 by H-bonding and limits access of large molecules.
- Proximal His (F8) transmits movement of Fe^{2+} to the F-helix during ligand binding.
- Small structural changes in this pocket are amplified to the whole protein (basis of regulation and cooperativity in hemoglobin).

6. Key Residues Forming the Heme Pocket

Helix	Residues (Example: Myoglobin)	Role
B	Leu B10	Hydrophobic contact with heme
E	Val E11, Leu E10, His E7 (distal)	Shape pocket, stabilize O_2 binding
F	His F8 (proximal), Leu F4	Coordinate Fe^{2+} , transmit signal
G	Val G11	Hydrophobic support
CD loop	Phe CD1, Leu CD3	Hydrophobic wall of pocket

7. Ligands that Bind at the Heme Iron (6th site)

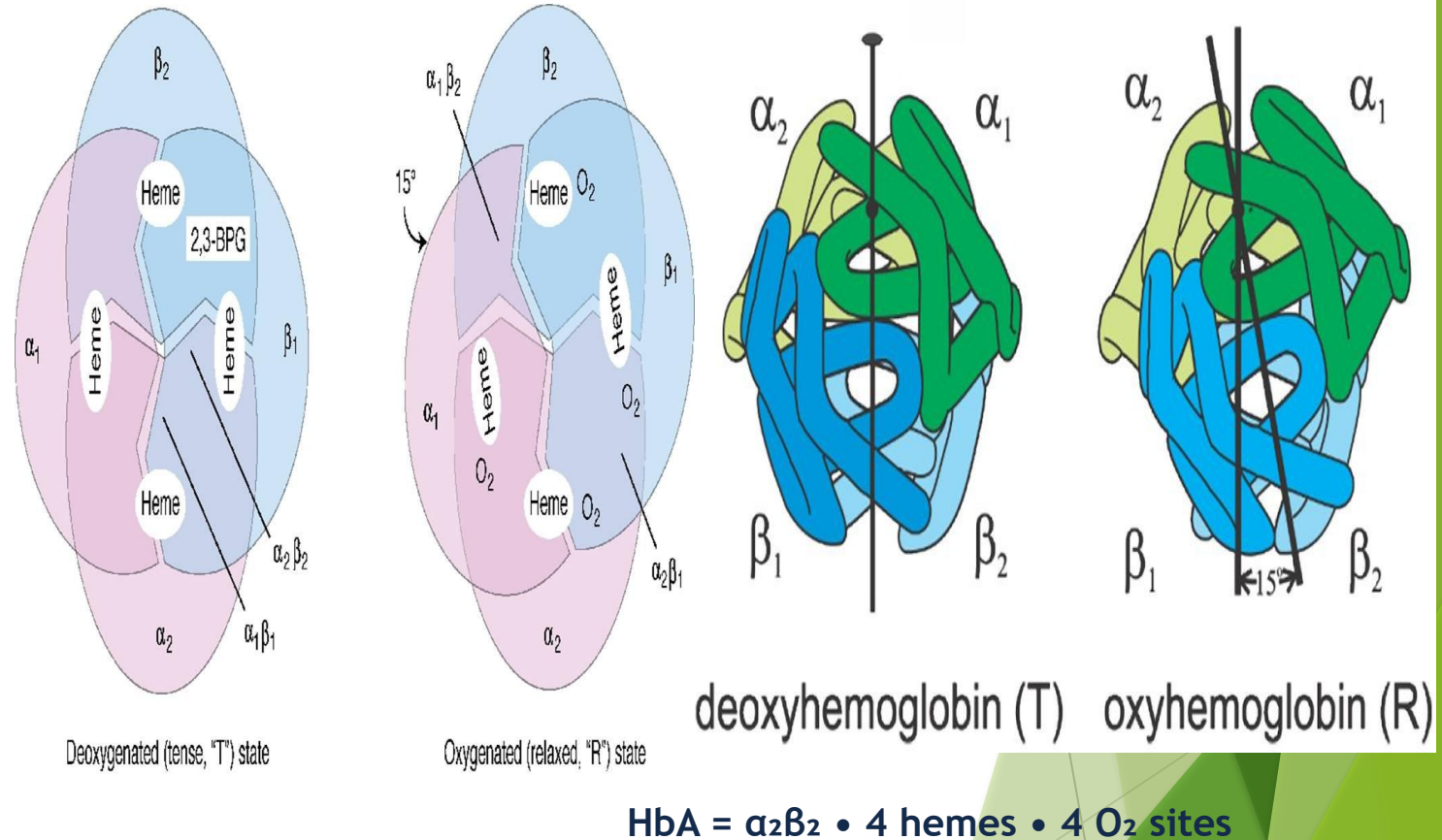


8. Summary

- The tertiary structure creates a buried hydrophobic heme pocket.
- Precise positioning of key residues (**His E7** and **His F8**) controls ligand binding.
- The heme pocket is the structural and functional heart of globins.

Quaternary structure and allostery

- Quaternary structure describes the association of two or more polypeptide chains.
- Hemoglobin is a heterotetramer: two α chains + two β chains in HbA.
- The $\alpha\beta$ pairs form two functional $\alpha\beta$ dimers that interact with one another.
- Interactions between subunits allow ligand binding at one site to influence the affinity of other sites.
- This inter-subunit communication is the structural basis of cooperativity and allostery.



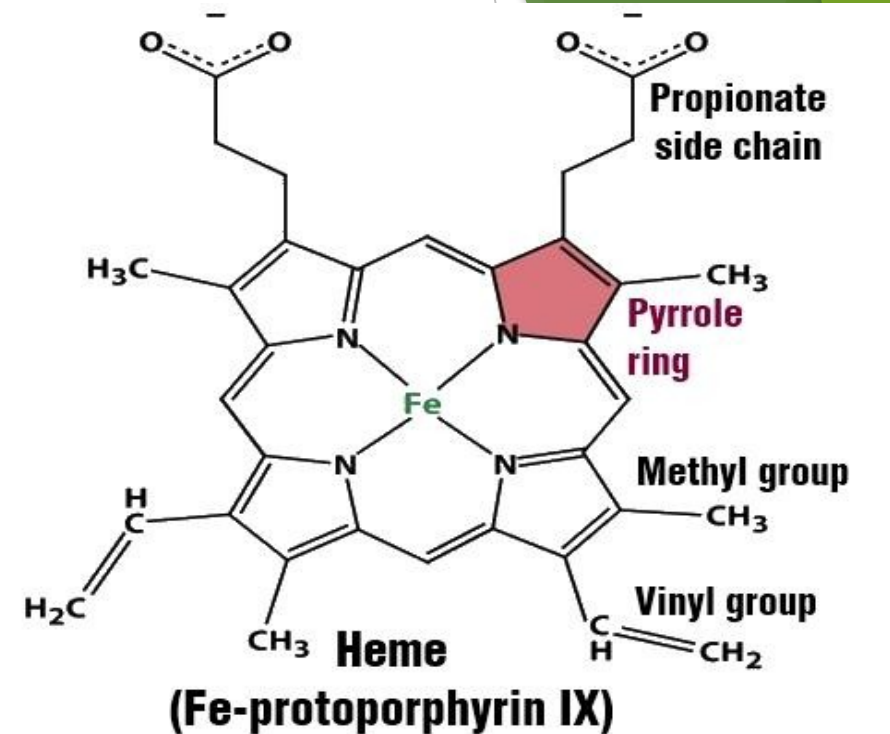
❖ *Myoglobin is **monomeric**; hemoglobin is **tetrameric**. This difference is central to **their different oxygen-binding behavior**.*

Part II — Heme & Myoglobin

Molecular structure of the oxygen-binding unit

Heme: The prosthetic group

- ❖ **Heme** is a prosthetic group found in several biologically important proteins, including **hemoglobin, myoglobin, cytochromes, catalase, and peroxidases**
- Heme is a prosthetic group consisting of **ferrous iron (Fe^{2+})** coordinated within **protoporphyrin IX**.
- Protoporphyrin IX contains four **pyrrole rings** connected by methine ($-\text{CH}=\text{}$) bridges, forming a planar conjugated porphyrin system.
- Substituents include methyl, vinyl and propionate groups.
- The conjugated ring system contributes strongly to the characteristic color of heme proteins.
- Fe^{2+} provides the reversible binding site for O_2 in functional myoglobin and hemoglobin.



Heme Structure

❖ Six-coordinate iron

- ❖ Four coordination positions come from pyrrole nitrogens; the fifth is the proximal histidine; the sixth is available for O_2 or another ligand.

Heme Coordination Chemistry

Coordination position	Ligand / role	Functional importance
1–4	Four pyrrole-ring nitrogens	Hold Fe ²⁺ in the porphyrin plane
5	Proximal histidine (F8)	Direct Fe ²⁺ –protein coordination
6	O ₂ in oxy-Mb/Hb	Reversible oxygen binding
Alternative 6th ligand	H ₂ O in methemoglobin	Fe ³⁺ state cannot bind O ₂ normally

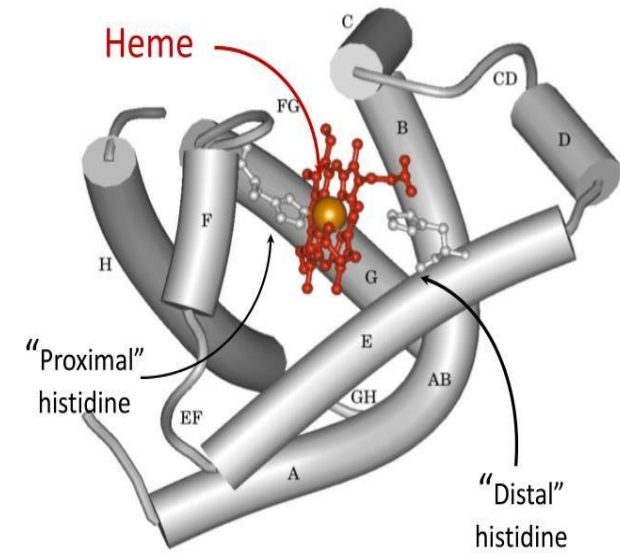
❖ Important distinction

- Proximal histidine directly coordinates iron. Distal histidine is on the opposite side of the heme pocket and does not directly ligate Fe.

Myoglobin: Overall Structure

- Myoglobin (Mb) is a small, monomeric heme protein found predominantly in **skeletal and cardiac muscle**.
- It contains **one globin polypeptide** and **one heme group**, so one Mb molecule can bind one O₂ molecule.
- About 80% of the polypeptide chain is organized into eight α -helical segments, conventionally labeled A–H.
- The heme lies in a **hydrophobic pocket**, while charged residues are largely solvent-exposed.
- **Physiologic role:** facilitate intracellular O₂ storage and diffusion toward mitochondria, especially when muscle oxygen demand rises.

Structure of Myoglobin



❖ *One Mb molecule = one heme = one O₂-binding site.*

Myoglobin: Ligand Binding Mechanism

- Deoxymyoglobin contains Fe^{2+} slightly displaced from the porphyrin plane toward the proximal histidine.
- When O_2 binds at the sixth coordination position, Fe^{2+} moves closer to the porphyrin plane.
- This movement is coupled to small shifts in the proximal histidine and nearby globin residues.
- Because Mb has one binding site and no interacting subunits, its O_2 binding is noncooperative.
- The result is a hyperbolic oxygen-binding curve.

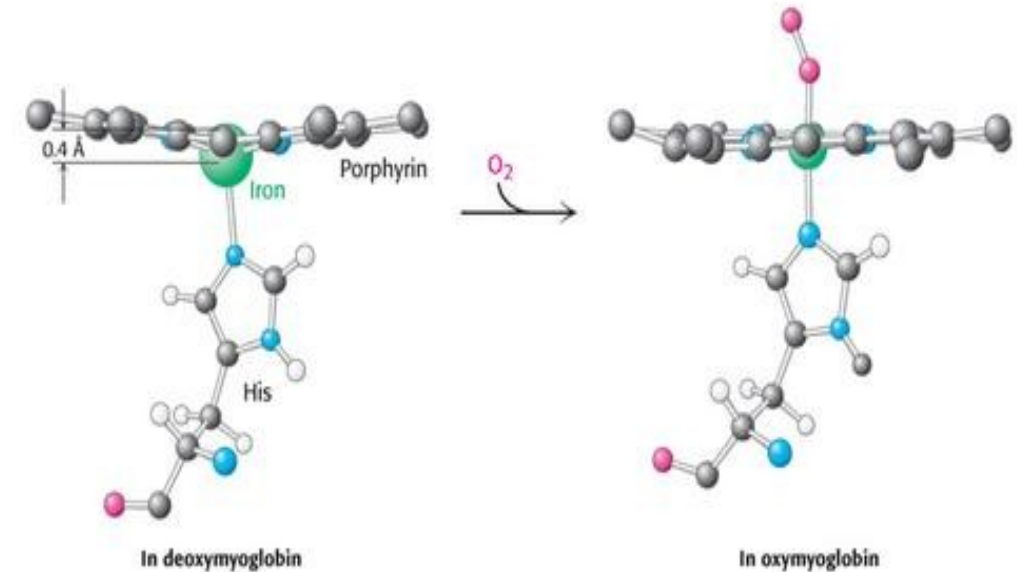


FIGURE 7.2 Oxygen binding changes the position of the iron ion. The iron ion lies slightly outside the plane of the porphyrin in deoxymyoglobin heme (left), but moves into the plane of the heme on oxygenation (right).

Myoglobin vs hemoglobin

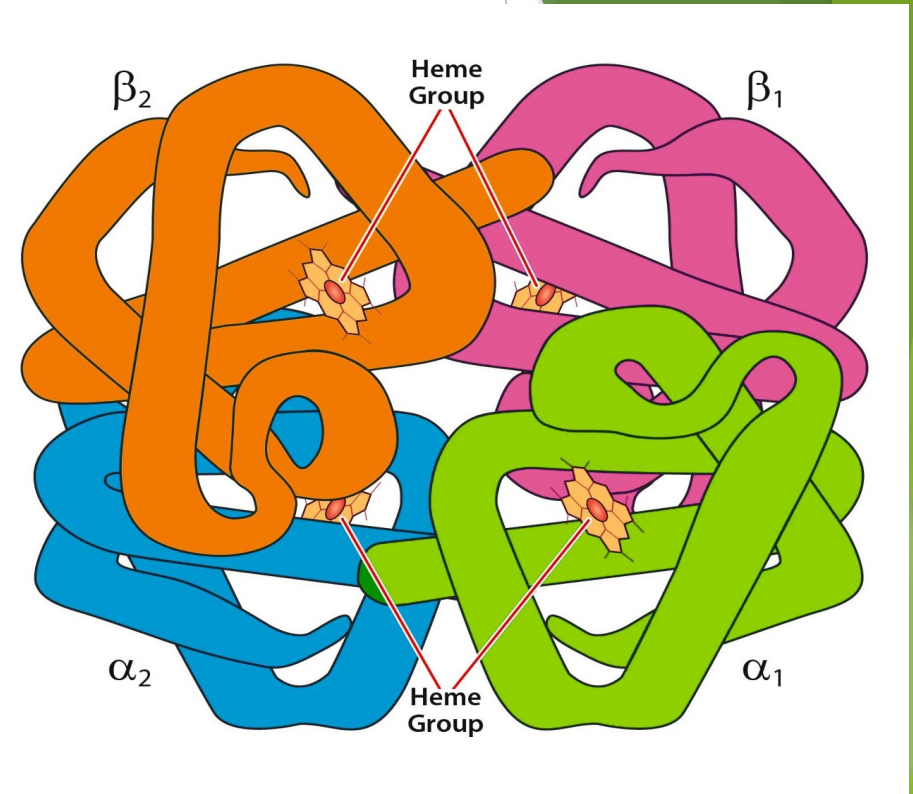
Feature	Myoglobin	Hemoglobin
Location	Muscle cells	Erythrocytes
Subunit organization	Monomer	Tetramer
Heme groups	1	4
O ₂ molecules bound	1	Up to 4
O ₂ -binding curve	Hyperbolic	Sigmoidal
Cooperativity	Absent	Present
Main role	Intracellular O ₂ storage/facilitation	O ₂ transport and CO ₂ /H ⁺ transport
P50 (approx.)	~2–3 mmHg in common reference data	~26–27 mmHg for adult Hb at standard conditions

Part III — Hemoglobin

Tetrameric structure, cooperativity and oxygen transport

Hemoglobin A: Molecular Organization

- ❖ HbA is the predominant adult hemoglobin and has the composition $\alpha_2\beta_2$.
- ❖ Each α chain contains 141 amino acids; each β chain contains 146 amino acids.
- ❖ Four heme groups are embedded in four globin subunits.
- ❖ Each heme provides one potential O_2 -binding site; therefore one Hb tetramer can bind **four O_2 molecules**.
- ❖ Subunits interact through hydrophobic contacts, ionic interactions and hydrogen bonds.
- ❖ These interfaces are dynamic and change during oxygenation and deoxygenation.



❖ **HbA = $\alpha_2\beta_2$ • 4 hemes • 4 O_2 sites**

❖ The key innovation of Hb is not merely carrying four O_2 molecules—it is communicating between four binding sites.

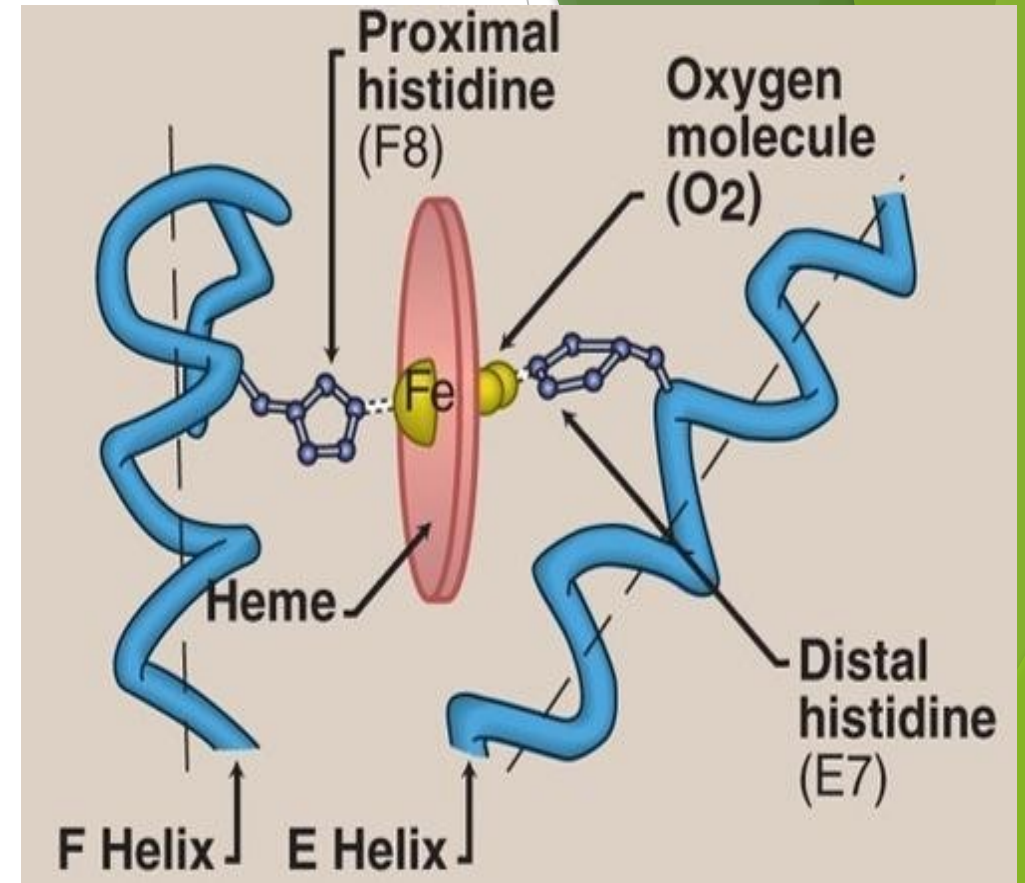
The Globin Chains: α and β

Chain	Length	HbA contribution	Important structural note
α -globin	141 aa	2 copies	Pair with β chains to form $\alpha\beta$ dimers
β -globin	146 aa	2 copies	Contains the HBB gene product; $\beta 6$ is clinically important in HbS
HbA total	574 aa	$\alpha_2\beta_2$	4 hemes; 4 potential O ₂ sites

- Globin chains adopt closely related folds, allowing them to interact with heme in a similar manner.
- The $\alpha 1\beta 1$ and $\alpha 2\beta 2$ interfaces are relatively stable, whereas $\alpha 1\beta 2$ and $\alpha 2\beta 1$ interfaces undergo major rearrangements during T $\leftrightarrow$ R transitions.

Proximal and Distal Histidines

- ❖ Proximal histidine (His F8) forms the fifth coordination bond to Fe^{2+} .
 - In myoglobin: His93 (F8)
 - In hemoglobin: α -chain: His87, β -chain: His92
- ❖ Distal histidine (His E7) lies on the opposite side of the heme and helps orient bound O_2 .
 - In myoglobin: His64 (E7)
 - In hemoglobin: α -chain: His58, β -chain: His63
- ❖ The distal pocket also helps reduce unwanted binding geometry and limits oxidation-related chemistry.
- ❖ Movement of Fe^{2+} during oxygenation pulls on the proximal histidine and the F-helix, contributing to the larger quaternary change.



❖ Structure → function

- A tiny atomic displacement at Fe^{2+} is amplified through the globin interfaces into a tetramer-wide conformational change.

T-state and R-state

Feature	T state (tense)	R state (relaxed)
O ₂ affinity	Lower	Higher
Typical ligand state	Deoxy / partially deoxygenated	Oxy / more oxygenated
Central cavity	Relatively larger	Smaller
Salt bridges	More stabilized	Many disrupted
2,3-BPG binding	Favored	Disfavored
Physiologic significance	Promotes O ₂ unloading	Promotes O ₂ loading

❖ *T and R are conformational ensembles, not two rigid shapes. Ligand binding shifts the equilibrium between them.*

Oxygenation: what changes structurally?

- ❖ Fe^{2+} is displaced slightly out of the porphyrin plane in deoxyhemoglobin.
 - When O_2 binds to the sixth coordination position:
 - $\text{Hb-Fe}^{2+} + \text{O}_2 \rightleftharpoons \text{Hb-Fe}^{2+} - \text{O}_2$
 - The O_2 binding pulls Fe^{2+} atom moves toward the plane of the porphyrin ring.
- ❖ The proximal histidine and attached F-helix move with the iron.
- ❖ These local changes alter $\alpha 1 \beta 2$ and $\alpha 2 \beta 1$ interfaces.
- ❖ Some ionic interactions and salt bridges are broken or rearranged.
- ❖ The tetramer shifts toward the R state, increasing the affinity of the remaining sites for O_2 .
- Oxygen Binding $\rightarrow$ Fe^{2+} movement $\rightarrow$ His/F-helix movement $\rightarrow$ interface arrangement $\rightarrow$ T $\rightarrow$ R $\rightarrow$ increased O_2 affinity

Part IV — Oxygen Binding & Transport

From molecular affinity to tissue oxygen delivery

Oxygen Dissociation Curves

- The oxygen dissociation curve plots hemoglobin or myoglobin saturation (Y) against the partial pressure of oxygen (pO_2).
- The partial pressure of oxygen needed to achieve half-saturation (P50) of the oxygen binding sites of myoglobin is approximately **1 mm Hg** and **26 mm Hg** for hemoglobin
- P50 is the pO_2 at which the protein is 50% saturated.
- Lower P50 = higher O_2 affinity; higher P50 = lower O_2 affinity.
- Mb shows a hyperbolic curve because it has one independent binding site.
- Hb shows a sigmoidal curve because binding at one site changes the affinity of other sites.

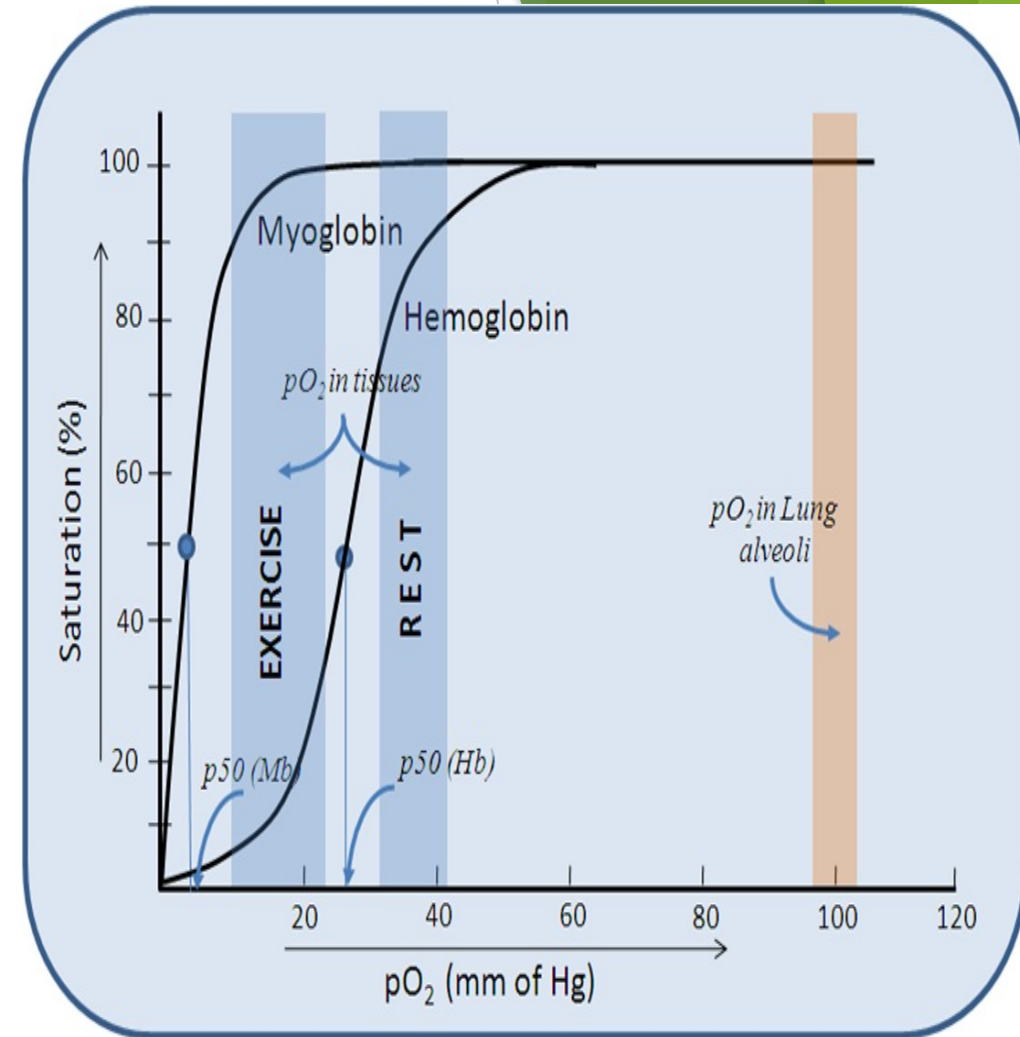
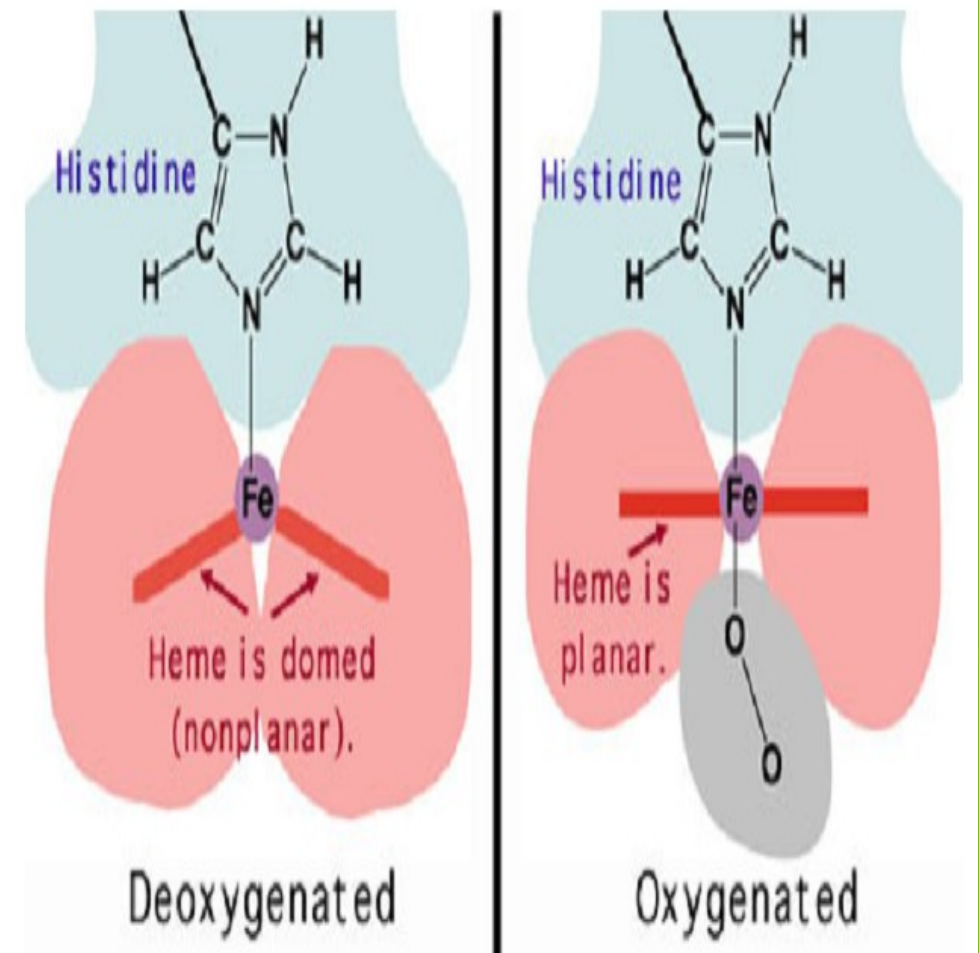


Figure.7.Oxygen dissociation curves for myoglobin and hemoglobin (Hb).

Cooperative oxygen binding

- ❖ Hemoglobin's four hemes are functionally coupled through subunit interactions.
- ❖ Binding of the first O_2 is relatively difficult because deoxyhemoglobin strongly favors the T state.
- ❖ Each subsequent O_2 binding event favors the R state and increases the affinity of remaining sites.
- ❖ This positive cooperativity produces the characteristic sigmoidal curve.
- ❖ Cooperativity allows Hb to load O_2 efficiently in the lungs while still unloading substantial O_2 in tissues.



❖ Why it matters

- A noncooperative carrier would be less effective at simultaneously achieving high loading and regulated

Oxygen loading in lungs vs unloading in tissues

Site	Typical environment	Hb behavior	Result
Pulmonary capillary	High pO ₂ , lower CO ₂ , relatively higher pH	High O ₂ affinity / R state favored	O ₂ loading
Systemic tissues	Lower pO ₂ , higher CO ₂ , lower pH	Affinity reduced / T state favored	O ₂ unloading
Active muscle	Lower pO ₂ + higher temperature + more H ⁺ /CO ₂	Right-shifted curve	Enhanced O ₂ delivery

The curve is an integrated physiological response—not simply an intrinsic property of Fe²⁺.

OXYGEN LOADING IN LUNGS vs. UNLOADING IN TISSUES

Hemoglobin is an allosteric protein. It binds O_2 efficiently in the lungs and releases O_2 in the tissues in response to local physiological conditions.

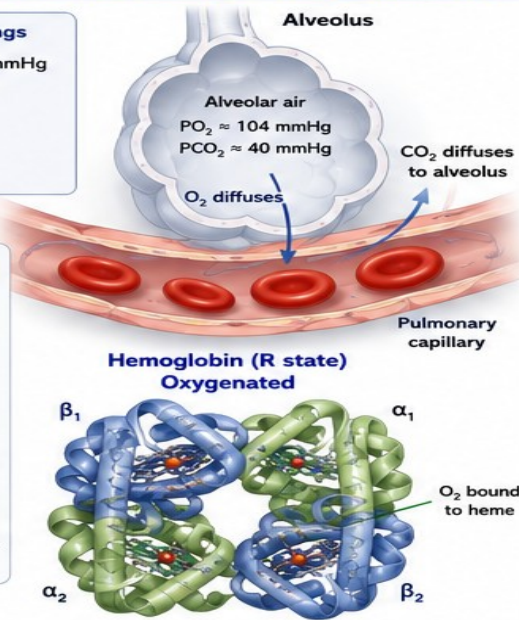
1. OXYGEN LOADING IN THE LUNGS (Pulmonary Capillaries)

Environment in the Lungs

High $pO_2 \approx 95-100$ mmHg
 Low $pCO_2 \approx 40$ mmHg
 High pH ≈ 7.40
 Lower temperature
 [2,3-BPG] has little effect

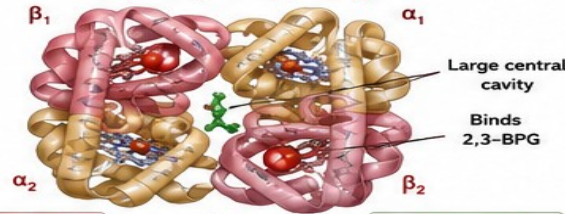
Oxygen Loading

- O_2 diffuses from alveolus into plasma and RBC.
- O_2 binds to Fe^{2+} of heme (6th coordination site).
- Fe^{2+} moves into the plane of the porphyrin.
- Proximal His (F8) moves $\rightarrow$ F-helix shifts.
- Subunit interfaces rearrange: T state $\rightarrow$ R state.
- Remaining heme sites show higher affinity and bind O_2 easily (positive cooperativity).



2. HEMOGLOBIN STRUCTURE AND CONFORMATIONAL CHANGE

T (Tense) State (Deoxy)



Factors that favor T state (Lower O_2 affinity)

$\downarrow pO_2$
 $\uparrow pCO_2$
 $\downarrow pH$ ($\uparrow H^+$)
 $\uparrow$ Temperature
 $\uparrow$ 2,3-BPG

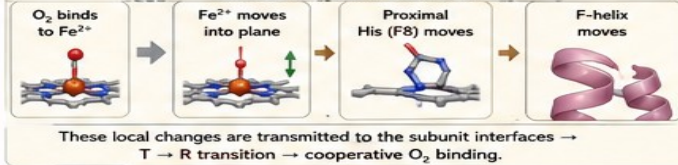
Factors that favor R state (Higher O_2 affinity)

$\uparrow pO_2$
 $\downarrow pCO_2$
 $\uparrow pH$ ($\downarrow H^+$)
 $\downarrow$ Temperature
 $\downarrow$ 2,3-BPG

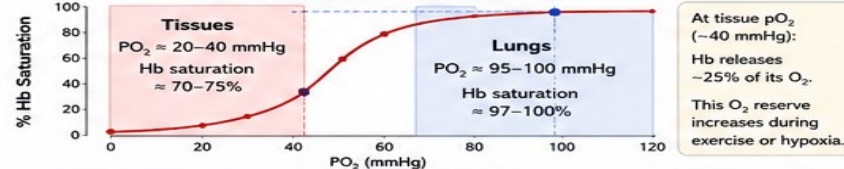
R (Relaxed) State (Oxy)



Key structural event when O_2 binds



4. OXYGEN DISSOCIATION CURVE



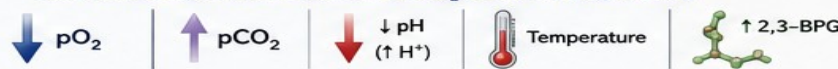
RESULT: O_2 LOADING + CO_2 UNLOADING

- Hb is mostly in the high-affinity R state.
- Arterial Hb saturation $\approx 97-100\%$.

PHYSIOLOGICAL ADVANTAGE

- Hemoglobin's allosteric properties allow it to adapt to local metabolic needs.
- High affinity in lungs ensures maximal loading.
- Low affinity in tissues ensures adequate delivery.

KEY FACTORS THAT DECREASE Hb- O_2 AFFINITY (Right Shift)



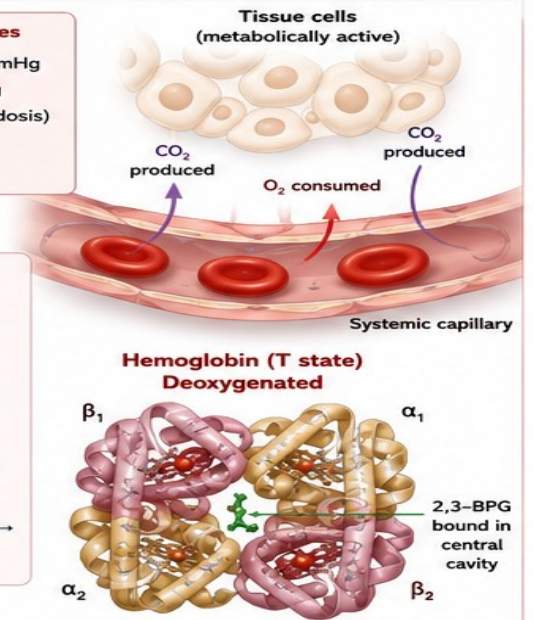
3. OXYGEN UNLOADING IN TISSUES (Systemic Capillaries)

Environment in Tissues

Low $pO_2 \approx 20-40$ mmHg
 High $pCO_2 \approx 46$ mmHg
 Low pH ≈ 7.20 (acidosis)
 Higher temperature
 $\uparrow$ 2,3-BPG (in RBC)

Oxygen Unloading

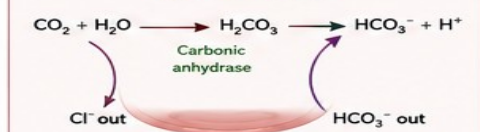
- Low pO_2 in tissues favors dissociation of O_2 from Hb.
- CO_2 diffuses into RBC.
- $CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic anhydrase).
- $H_2CO_3 \rightarrow HCO_3^- + H^+$.
- $\uparrow H^+$ ($\downarrow pH$) and $\uparrow CO_2$ + $\uparrow$ temperature + 2,3-BPG stabilize T state.
- Hb affinity for O_2 decreases $\rightarrow O_2$ is released to tissues.



CO_2 Loading (Reverse Haldane)

- Deoxygenated Hb binds H^+ and CO_2 more readily.
- HCO_3^- leaves RBC (Cl^- shift) into plasma.
- CO_2 is carried as HCO_3^- , carbamino-Hb, and dissolved CO_2 .

In RBC in Tissues



RESULT: O_2 UNLOADING + CO_2 LOADING

- Hb is mostly in the low-affinity T state.
- Mixed venous Hb saturation $\approx 70-75\%$ (at rest).

CLINICAL CORRELATES

- Acidosis, fever, anemia, high altitude, exercise $\rightarrow$ right shift $\rightarrow$ more O_2 unloading.
- Alkalosis, hypothermia, low 2,3-BPG, fetal Hb $\rightarrow$ left shift $\rightarrow$ less O_2 unloading.

Oxygen content of blood

- ❖ Most blood O_2 is carried bound to hemoglobin; only a small fraction is physically dissolved in plasma.
- ❖ The commonly used clinical approximation for O_2 bound to Hb is: Hb-bound $O_2 \approx 1.34 \times \text{Hb concentration} \times \text{fractional saturation}$.
- ❖ Example: at $\text{Hb} = 15 \text{ g/dL}$ and 100% saturation, Hb-bound $O_2 \approx 1.34 \times 15 \approx 20.1 \text{ mL } O_2/\text{dL}$.
- ❖ Dissolved O_2 contributes little to total content but determines pO_2 and therefore drives diffusion.
- ❖ Thus, anemia can markedly reduce O_2 content even when arterial pO_2 and saturation are near normal.

❖ Clinical distinction

- pO_2 measures dissolved oxygen; saturation measures Hb occupancy; oxygen content depends strongly on hemoglobin concentration.

Part V — Allosteric Regulation

2,3-BPG, Bohr effect, temperature and chloride

2,3-Bisphosphoglycerate (2,3-BPG)

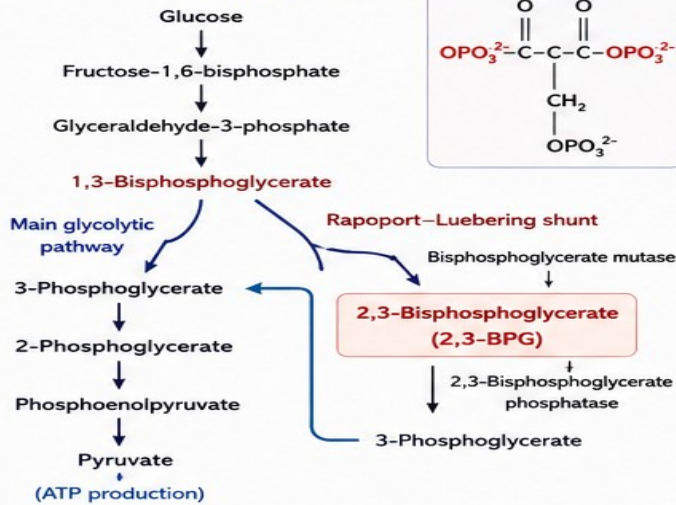
- ❖ 2,3-BPG is produced in erythrocytes from the glycolytic intermediate 1,3-bisphosphoglycerate through the Rapoport–Luebering shunt.
- ❖ It binds preferentially in the central cavity of deoxyhemoglobin.
- ❖ Positively charged residues in the cavity interact electrostatically with the negatively charged phosphate groups of 2,3-BPG.
- ❖ This stabilizes the T state and lowers hemoglobin's O₂ affinity.
- ❖ The physiological result is enhanced O₂ unloading in tissues.

2,3-BISPHOSPHOGLYCERATE (2,3-BPG): THE KEY ALLOSTERIC REGULATOR OF HEMOGLOBIN

Produced in erythrocytes via the Rapoport–Luebering shunt; binds in the central cavity of deoxyhemoglobin; stabilizes the T state; decreases Hb–O₂ affinity → enhances O₂ unloading in tissues.

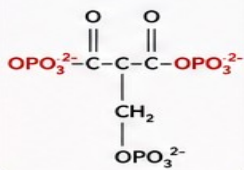
1. PRODUCTION OF 2,3-BPG IN ERYTHROCYTES (RAPOPORT–LUEBERING SHUNT)

2,3-BPG is formed from the glycolytic intermediate 1,3-bisphosphoglycerate.

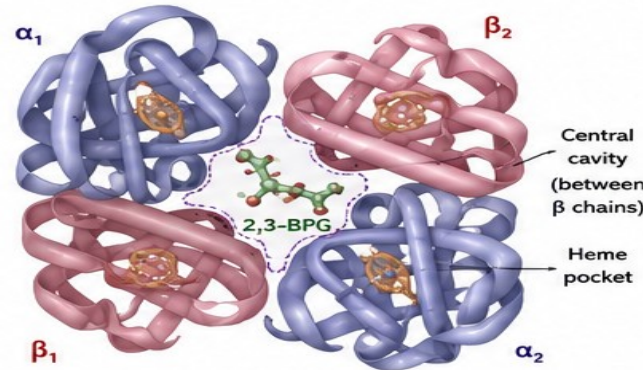


This shunt bypasses the ATP-generating steps of glycolysis and is active only in red blood cells (which lack mitochondria).

Structure of 2,3-BPG



2. BINDING SITE: THE CENTRAL CAVITY OF DEOXYHEMOGLOBIN (T STATE)



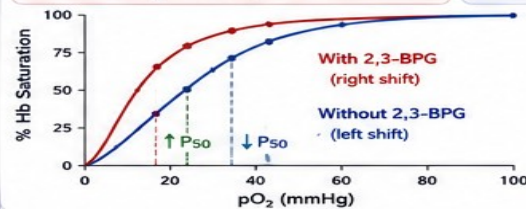
4. EFFECT ON HEMOGLOBIN AND OXYGEN AFFINITY

WITH 2,3-BPG (T state stabilized)

- T state is stabilized.
- Hemoglobin has lower affinity for O₂.
- O₂ dissociation curve shifts to the RIGHT.
- Promotes O₂ unloading in tissues.

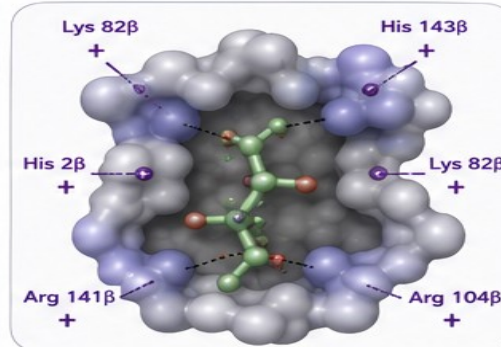
WITHOUT 2,3-BPG (R state favored)

- R state is favored.
- Hemoglobin has higher affinity for O₂.
- O₂ dissociation curve shifts to the LEFT.
- Reduces O₂ unloading.



P₅₀ = pO₂ at which Hb is 50% saturated.
 ↑ P₅₀ = lower affinity (right shift)
 ↓ P₅₀ = higher affinity (left shift)

3. MOLECULAR INTERACTIONS IN THE CENTRAL CAVITY



- ✓ The cavity contains positively charged residues (Lys, Arg, His).
- ✓ The negatively charged phosphate groups of 2,3-BPG interact electrostatically (salt bridges) with these residues.
- ✓ These interactions stabilize the deoxy (T) conformation.
- ✓ 2,3-BPG is expelled when Hb converts to the oxygenated (R) state.

2,3-BPG (negatively charged)
 ⊕ Positively charged residues
 ---- Electrostatic interactions

5. PHYSIOLOGICAL ROLE OF 2,3-BPG

Tissues
 (High CO₂, Low pH, Low pO₂)



Binds deoxy Hb (T state)
 ↓ Hb–O₂ affinity
 ↑ O₂ unloading

Low 2,3-BPG (stored blood, liver disease) → left shift → impaired O₂ unloading → tissue hypoxia.

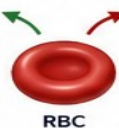
Importance

- ✓ Matches O₂ delivery with metabolic demand.
- ✓ Enhances O₂ release in active tissues.
- ✓ High 2,3-BPG levels:
 - Anemia
 - Chronic hypoxia
 - High altitude
 - Exercise

6. REGULATION OF 2,3-BPG LEVELS IN RBCs

Increased 2,3-BPG ↑

- ✓ Hypoxia (low O₂)
- ✓ Acidosis (↑ CO₂, ↑ H⁺)
- ✓ Chronic lung disease
- ✓ High altitude
- ✓ Anemia
- ✓ Exercise training



Decreased 2,3-BPG ↓

- ✓ Stored blood (> 2–3 weeks)
- ✓ Liver disease
- ✓ Congestive heart failure
- ✓ Newborns (low levels)
- ✓ After blood transfusion

7. SUMMARY

Feature	With High 2,3-BPG	Without 2,3-BPG
Binding site	Central cavity (β chains)	No binding
Hb conformation	T state stabilized	R state favored
O ₂ affinity	Decreased	Increased
P ₅₀	Increased	Decreased
O ₂ dissociation curve	Right shift	Left shift
Physiological outcome	Enhanced O ₂ unloading	Reduced O ₂ unloading

8. KEY TAKE-HOME MESSAGE



High 2,3-BPG → Low O₂ affinity → More O₂ released to tissues.

2,3-BPG is a master regulator that tunes hemoglobin's oxygen affinity to ensure adequate oxygen delivery from lungs to tissues.

2,3-BPG: physiological and clinical relevance

Condition	Typical 2,3-BPG response	Functional consequence
High altitude / chronic hypoxemia	↑	Facilitates tissue O ₂ unloading
Chronic anemia	↑	Helps compensate for reduced Hb mass
Some chronic hypoxic states	↑	Right shift; lower Hb affinity
Stored blood	↓ with storage	Can transiently reduce O ₂ unloading after transfusion; levels recover after transfusion

❖ *The exact time course and magnitude of 2,3-BPG changes depend on the clinical setting and duration of hypoxia/anemia.*

Bohr effect: CO₂ and H⁺

- The Bohr effect describes the reduction in hemoglobin O₂ affinity caused by increased H⁺ concentration (lower pH) and, physiologically, increased CO₂.
- Metabolically active tissues generate CO₂ and acids, promoting O₂ unloading.
- Protonation of selected globin residues favors the deoxy/T state.
- CO₂ can form carbamates with amino-terminal groups of globin chains, generating additional charge changes that stabilize deoxyhemoglobin.
- The oxygen dissociation curve shifts to the right: P₅₀ increases.

❖ One-line summary

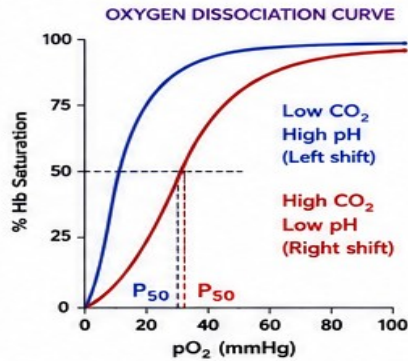
➤ $\uparrow\text{CO}_2 + \uparrow\text{H}^+ \rightarrow \downarrow\text{Hb-O}_2 \text{ affinity} \rightarrow \uparrow\text{O}_2 \text{ unloading.}$

BOHR EFFECT: CO₂ AND H⁺

Increase in CO₂ and H⁺ (decrease in pH) decreases hemoglobin's affinity for O₂, promoting O₂ unloading in tissues.

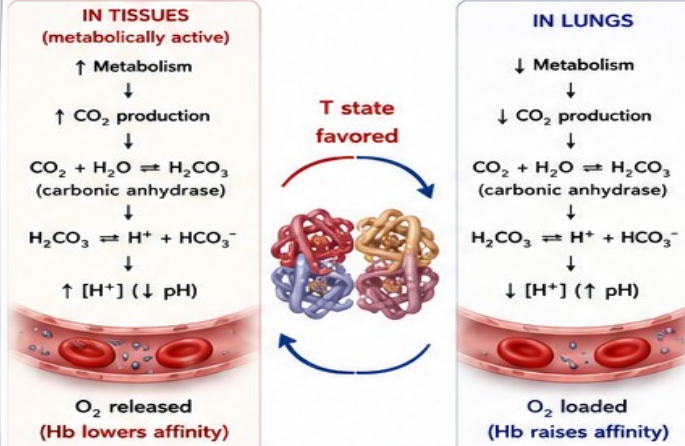
1. WHAT IS THE BOHR EFFECT?

- ✓ The Bohr effect describes the reduction in hemoglobin (Hb) O₂ affinity caused by increased H⁺ concentration (lower pH) and, physiologically, increased CO₂.
- ✓ It is a key mechanism that ensures O₂ is released where metabolism is active.
- ✓ Oxygen dissociation curve shifts to the **RIGHT**.
- ✓ **Result: P₅₀ increases** (Hb has lower affinity for O₂).

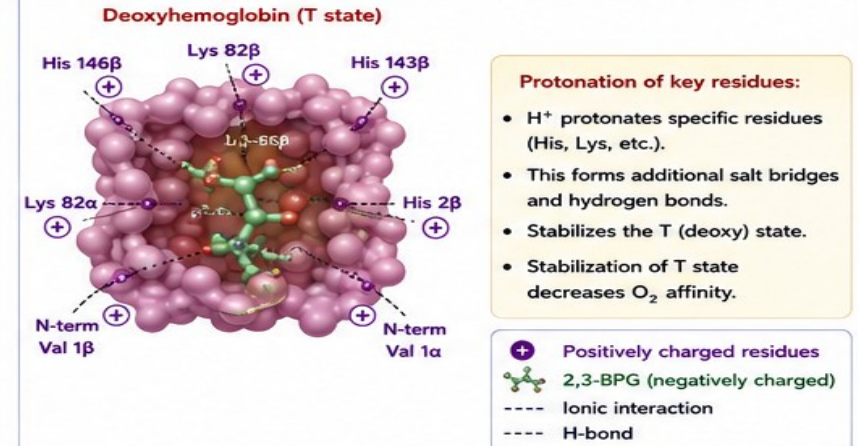


Right shift → P₅₀ ↑ → ↓ O₂ affinity → ↑ O₂ unloading

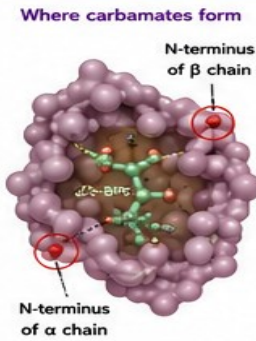
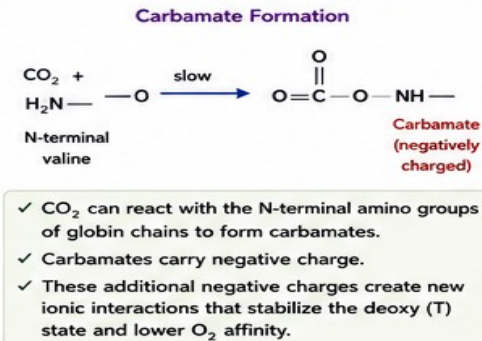
2. WHERE AND WHY DOES IT OCCUR?



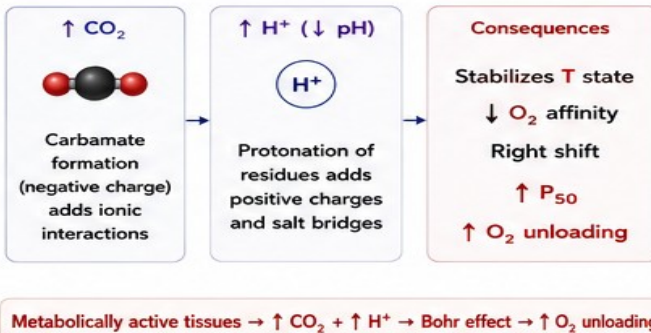
3. HOW H⁺ (LOW pH) REDUCES O₂ AFFINITY



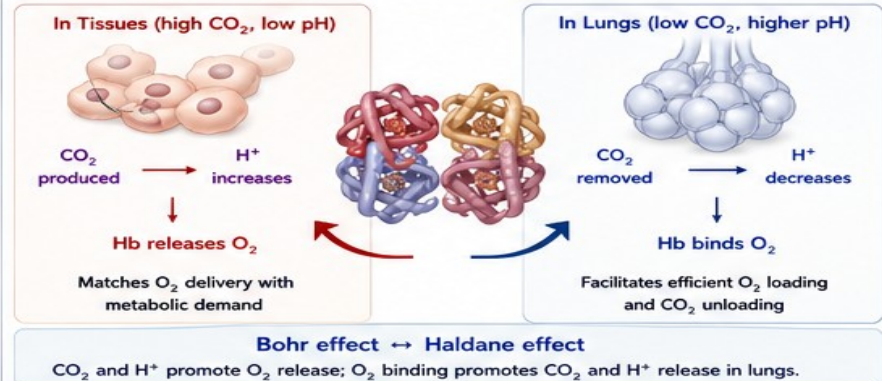
4. ROLE OF CO₂



5. COMBINED EFFECT OF CO₂ AND H⁺



6. PHYSIOLOGICAL IMPORTANCE



SUMMARY

- ✓ ↑ CO₂ and ↑ H⁺ (↓ pH) decrease hemoglobin's affinity for O₂.
- ✓ This stabilizes the T state (deoxyhemoglobin).
- ✓ The O₂ dissociation curve shifts to the right.
- ✓ P₅₀ increases.
- ✓ Physiological result: enhanced O₂ unloading in tissues.

EFFECTS ON OXYGEN DISSOCIATION CURVE

Condition	pH / H ⁺	CO ₂	Curve Shift	P ₅₀	Hb-O ₂ Affinity	O ₂ Unloading
↑ CO ₂ , ↑ H ⁺ (↓ pH)	Low	High	Right	↑ Increase	↓ Decreased	↑ Increased
↓ CO ₂ , ↓ H ⁺ (↑ pH)	High	Low	Left	↓ Decrease	↑ Increased	↓ Decreased

KEY TAKE-HOME MESSAGE



The Bohr effect is an elegant physiological mechanism that links cellular metabolism to oxygen delivery—ensuring oxygen is released where it is needed most.

Carbon dioxide transport

Form	Approximate contribution	Biochemical route
Dissolved CO ₂	~5–10%	Physically dissolved in plasma
Carbaminohemoglobin	~10–20%	CO ₂ reacts with amino-terminal groups of globin chains
Bicarbonate (HCO ₃ ⁻)	~70–80%	CO ₂ + H ₂ O ⇌ H ₂ CO ₃ ⇌ H ⁺ + HCO ₃ ⁻ ; carbonic anhydrase accelerates the reaction in RBCs

❖ *Percentages vary with physiological conditions; bicarbonate is the major transport form of CO₂.*

Carbonic anhydrase and the chloride shift

- ❖ In systemic tissues, CO_2 enters erythrocytes and is rapidly hydrated by carbonic anhydrase.
- ❖ H_2CO_3 dissociates to $\text{H}^+ + \text{HCO}_3^-$.
- ❖ Hemoglobin buffers the generated H^+ , helping sustain CO_2 conversion.
- ❖ Bicarbonate exits the RBC in exchange for chloride (Cl^-): the chloride shift.
- ❖ In the lungs, the process reverses: bicarbonate re-enters RBCs, CO_2 is regenerated and then exhaled.

❖ *Carbonic anhydrase is central to rapid $\text{CO}_2/\text{HCO}_3^-$ interconversion in erythrocytes.*

Haldane effect

- ❖ The Haldane effect describes how deoxygenated hemoglobin carries CO_2 and H^+ more effectively than oxygenated hemoglobin.
- ❖ In systemic tissues, deoxygenation of Hb promotes H^+ buffering and CO_2 carriage.
- ❖ In the lungs, oxygenation of Hb reduces its affinity for H^+ and CO_2 , facilitating their release.
- ❖ The Bohr effect mainly describes how CO_2/H^+ affect O_2 binding; the Haldane effect describes how O_2 binding affects CO_2/H^+ transport.
- ❖ Together they couple pulmonary gas exchange with systemic metabolism.

❖ Complementary effects

➤ Bohr: $\text{CO}_2/\text{H}^+ \rightarrow \text{O}_2$ unloading. Haldane: O_2 loading $\rightarrow \text{CO}_2/\text{H}^+$ unloading.

Temperature and chloride effects

Factor	Change	Curve effect	Physiologic setting
Temperature	↑	Right shift	Exercise, fever
Temperature	↓	Left shift	Hypothermia
Chloride	Binding to deoxy-Hb favors T state	Reduced O ₂ affinity	Supports CO ₂ transport / tissue unloading
pH	↓	Right shift	Active metabolism
pH	↑	Left shift	Pulmonary environment

The oxygen dissociation curve: right vs left shift

- ❖ Right shift = decreased O₂ affinity, higher P₅₀, enhanced unloading.
- ❖ Common causes: ↑H⁺/↓pH, ↑CO₂, ↑temperature and ↑2,3-BPG.
- ❖ Left shift = increased O₂ affinity, lower P₅₀, tighter O₂ binding.
- ❖ Common causes: ↓CO₂, ↑pH, ↓temperature, ↓2,3-BPG, HbF and CO exposure.
- ❖ Remember: a left shift does not necessarily mean improved tissue oxygenation; excessive affinity can impair unloading.

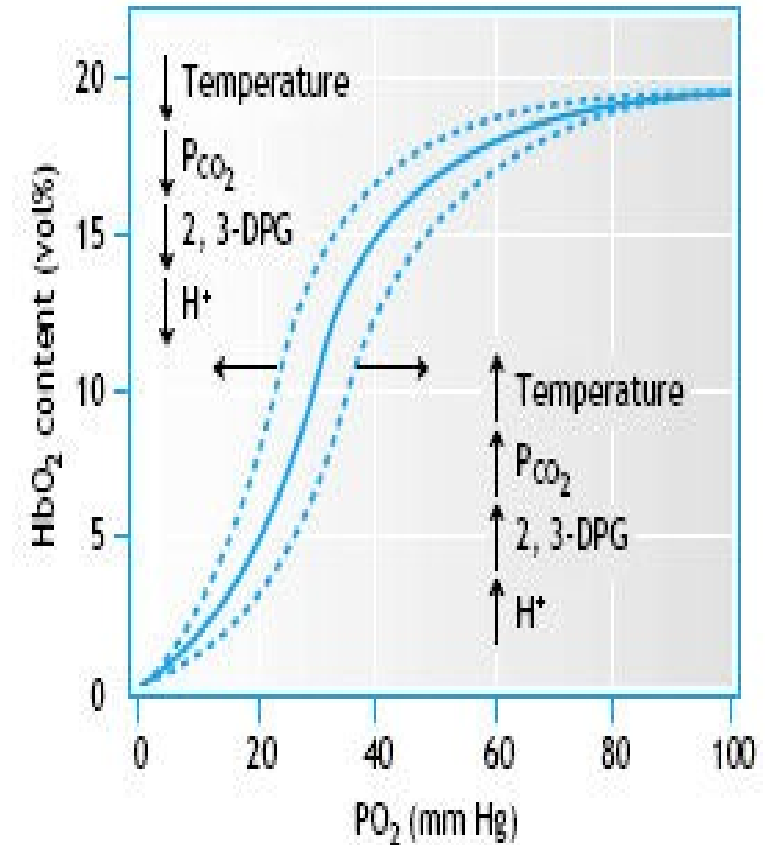


Figure 3-7. Changes in affinity of hemoglobin (Hb) for O₂ (oxyhemoglobin dissociation curve).

NB. Allosteric regulators favor the unloading of oxygen in the peripheral tissues

Part VI — Developmental Hemoglobins & Derivatives

Normal Variants And Clinically Important Forms

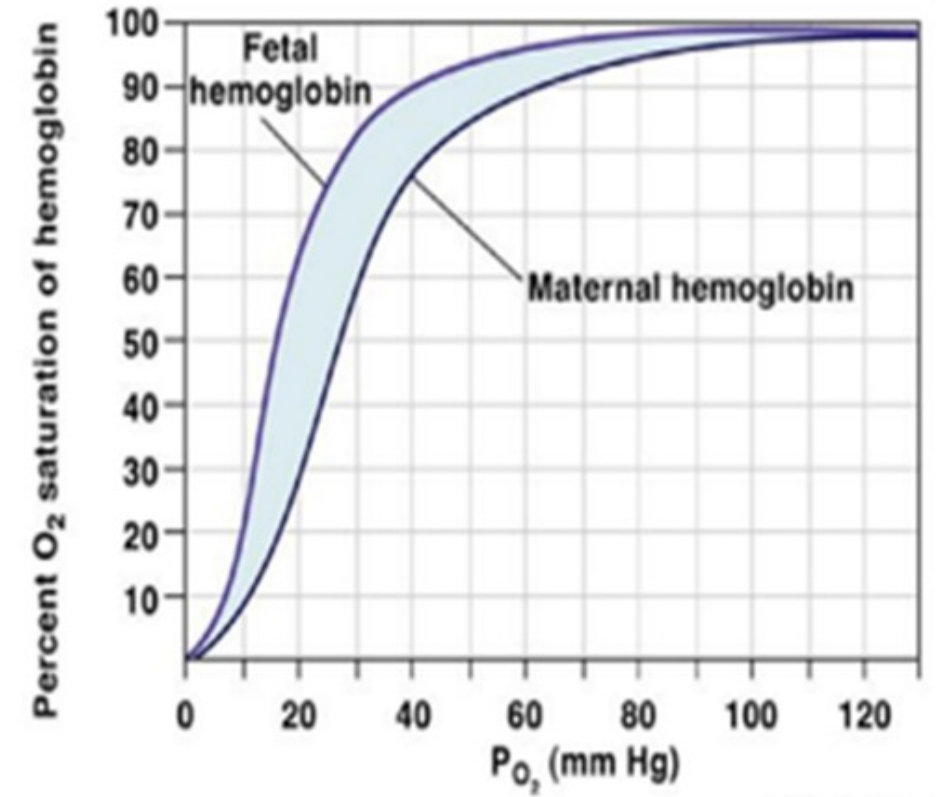
Adult, fetal and embryonic hemoglobins

Hemoglobin	Globin composition	Major period	Functional significance
HbA	$\alpha_2\beta_2$	Predominant after infancy	Normal adult Hb
HbA ₂	$\alpha_2\delta_2$	Minor adult fraction	Useful in evaluation of β -thalassemia
HbF	$\alpha_2\gamma_2$	Fetal life; small amount in adults	Higher O ₂ affinity; facilitates placental O ₂ transfer
Embryonic Hbs	ζ/ϵ combinations	Very early development	Developmental oxygen transport

❖ *Hb composition changes during development because globin-gene expression switches from embryonic → fetal → adult patterns.*

Why HbF binds oxygen more tightly

- HbF contains γ chains rather than β chains: $\alpha_2\gamma_2$.
- γ chains bind 2,3-BPG less strongly than β chains.
- Therefore HbF is less stabilized in the T state by 2,3-BPG.
- HbF has a left-shifted O_2 dissociation curve compared with HbA.
- This higher affinity supports oxygen transfer from maternal HbA to fetal HbF across the placenta.



Hemoglobin derivatives

Derivative	Iron / ligand state	Key feature	Clinical relevance
Oxyhemoglobin	$\text{Fe}^{2+} + \text{O}_2$	Normal oxygenated Hb	Normal O_2 carriage
Deoxyhemoglobin	Fe^{2+} , no O_2	T-state favored	Normal tissue unloading
Carboxyhemoglobin	$\text{Fe}^{2+} + \text{CO}$	CO binds strongly; increases affinity at remaining sites	Carbon monoxide poisoning
Methemoglobin	Fe^{3+}	Cannot bind O_2 normally	Methemoglobinemia
HbA1c	Glycated HbA	Stable nonenzymatic glycation product	Reflects average glycemia over prior weeks

Carbon monoxide poisoning

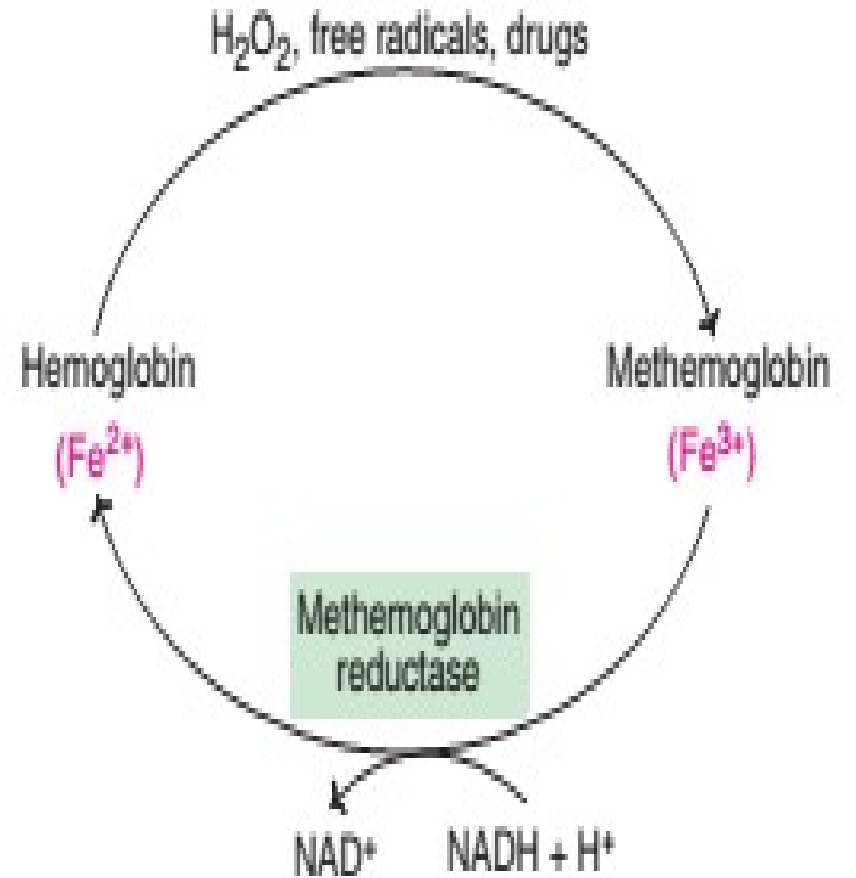
- Carbon monoxide (CO) competes with O₂ for heme binding and has a much higher affinity for Hb than O₂.
- CO binding forms carboxyhemoglobin (COHb).
- CO also shifts the remaining O₂-binding sites toward higher affinity, producing a left shift and impairing tissue unloading.
- The major pathophysiologic problem is reduced effective oxygen delivery plus cellular hypoxia.
- Pulse oximetry can be misleading because it does not reliably distinguish oxyhemoglobin from carboxyhemoglobin; co-oximetry is used for direct multi-wavelength measurement.

❖ Clinical pattern

- Severe CO exposure may produce headache, dizziness, confusion, loss of consciousness and tissue hypoxia; urgent clinical assessment is required.

Methemoglobin

- ❖ Functional Hb requires iron in the Fe^{2+} state.
- ❖ Oxidation converts Fe^{2+} to Fe^{3+} , producing methemoglobin.
- ❖ Methemoglobin cannot bind O_2 normally at the oxidized heme.
- ❖ Oxidative stress, certain drugs/chemicals and inherited defects can increase methemoglobin.
- ❖ Red blood cells normally contain enzymatic systems that continuously reduce small amounts of methemoglobin back toward functional Hb.



❖ *Key distinction: Fe^{3+} methemoglobin is an oxidation-state problem, not simply 'deoxygenated Hb'.*

HbA1c: structural chemistry

- ❖ HbA1c forms by nonenzymatic glycation of hemoglobin in circulating erythrocytes.
- ❖ Glucose-derived chemistry initially forms a reversible Schiff base and then a more stable ketoamine (Amadori product).
- ❖ A major HbA1c species involves the N-terminal valine of the β chain.
- ❖ Because erythrocytes circulate for roughly 120 days, HbA1c reflects average glycemic exposure over the preceding ~2–3 months, with greater weighting toward recent weeks.
- ❖ HbA1c interpretation can be affected by conditions that alter red-cell survival or hemoglobin composition.

❖ Biochemistry → clinical test

- The assay measures a modified protein, linking protein chemistry to long-term glucose monitoring.

Part VII — Hemoglobinopathies

How small sequence changes can produce major physiological effects

Hemoglobinopathies: overview

- Hemoglobinopathies are inherited disorders involving abnormal globin structure, quantity, or both.
- Structural variants arise from changes in globin amino-acid sequence.
- Thalassemias are quantitative disorders: synthesis of α - or β -globin is reduced or absent.
- Important examples include HbS, HbC, α -thalassemia and β -thalassemia.
- The clinical phenotype depends on the molecular defect, genotype, modifier genes and environmental/physiological factors.

Category	Primary defect	Example
Structural variant	Altered amino-acid sequence	HbS, HbC
Thalassemia	Reduced globin synthesis	α -thalassemia, β -thalassemia
Compound state	More than one variant/defect	HbS/ β -thalassemia

Sickle hemoglobin (HbS): molecular basis

- ❖ HbS results from a β -globin structural mutation in the HBB gene.
- ❖ The classic HbS substitution is $\beta 6 \text{ Glu} \rightarrow \text{Val}$: glutamic acid is replaced by valine at position 6 of the β chain.
- ❖ This replaces a charged, hydrophilic residue with a hydrophobic residue.
- ❖ When deoxygenated, HbS molecules can polymerize, producing long intracellular fibers.
- ❖ Polymerization distorts erythrocytes into rigid sickle forms, promoting hemolysis and microvascular obstruction.

Sickle cell disease: structure → phenotype

- ❖ Deoxygenated HbS has increased tendency to polymerize because hydrophobic interactions become exposed between molecules.
- ❖ Repeated cycles of sickling and unsickling damage the red-cell membrane.
- ❖ Consequences include chronic hemolytic anemia, vaso-occlusion, tissue ischemia and painful crises.
- ❖ HbF is an important modifier because increased HbF can reduce HbS polymerization.
- ❖ Laboratory diagnosis commonly relies on hemoglobin analysis (e.g., electrophoresis/HPLC) together with clinical and genetic information.

Hemoglobin C (HbC)

- HbC is another β -globin structural variant.
- The classic mutation is $\beta 6 \text{ Glu} \rightarrow \text{Lys}$, replacing an acidic residue with a positively charged basic residue.
- HbC has reduced solubility and can form intracellular crystals under appropriate conditions.
- HbC disease is often associated with hemolytic anemia of variable severity.
- HbC can occur with HbS or β -thalassemia, producing compound phenotypes.

❖ Compare with HbS

- Both HbS and HbC involve $\beta 6$, but the amino-acid substitution is different: $\text{Glu} \rightarrow \text{Val}$ in HbS versus $\text{Glu} \rightarrow \text{Lys}$ in HbC.

Thalassemias: quantitative globin defects

Type	Primary problem	Major consequence
α -thalassemia	Reduced/absent α -globin production	Imbalance of globin chains; microcytosis/hemolysis; severity depends on number of affected α genes
β -thalassemia	Reduced/absent β -globin production	Excess α chains, ineffective erythropoiesis and anemia; severity varies by genotype

- ❖ Unlike HbS/HbC, the central defect is usually the amount of globin produced rather than a single abnormal amino acid in a normal quantity of chain.
- ❖ Unbalanced globin-chain synthesis damages erythroid precursors and mature RBCs.
- ❖ Peripheral blood indices and hemoglobin analysis are important laboratory clues; molecular testing can define the genotype.

BY: Mabratu Takele Desisa

Part VIII — Pulmonary Surfactant

Molecular composition, alveolar mechanics and clinical relevance

Why alveoli need surfactant

- ❖ Alveoli are tiny air-filled structures where O_2 and CO_2 exchange occurs across a very thin blood–air barrier.
- ❖ The air–liquid interface creates surface tension that tends to reduce alveolar surface area and promote collapse.
- ❖ Pulmonary surfactant lowers surface tension, particularly at low lung volumes.
- ❖ This reduces the work of breathing and helps prevent end-expiratory alveolar collapse (atelectasis).
- ❖ Surfactant also participates in innate pulmonary defense.
- ❖ *Surfactant is a biochemical solution to a physical problem: surface tension at the alveolar air–liquid interface.*

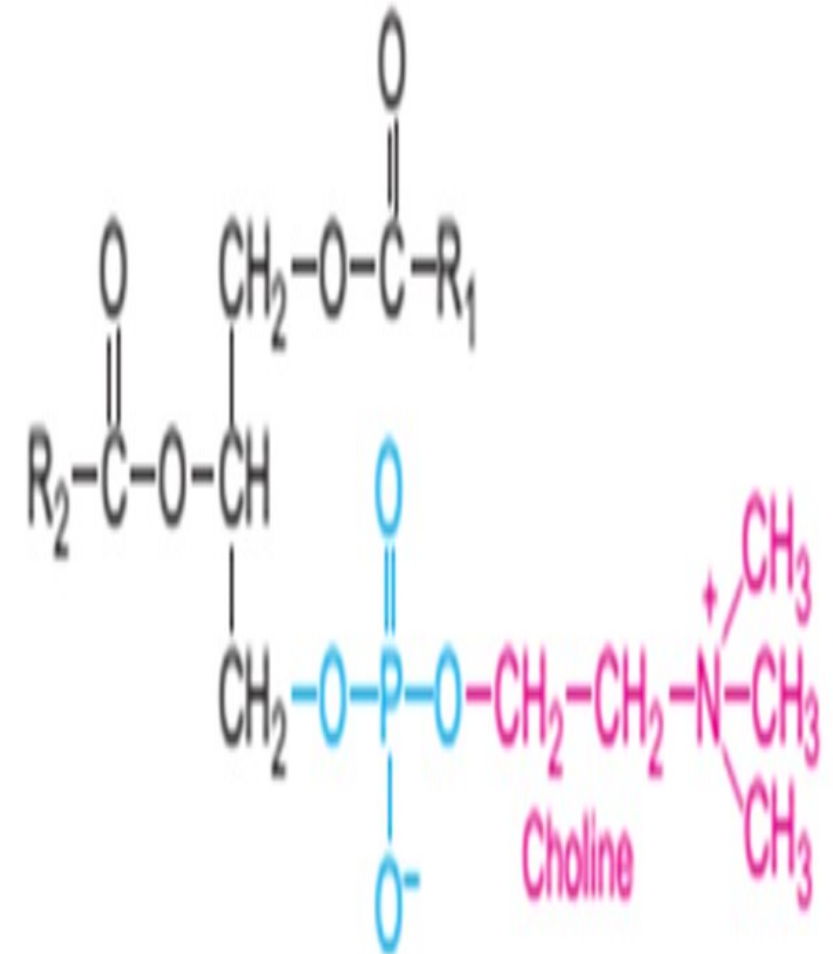
Pulmonary surfactant: composition

Component	Approximate role / abundance	Key function
Phospholipids	Majority of surfactant	Create the surface-active monolayer
DPPC	Major single phospholipid	Strong surface-tension lowering activity
Other phospholipids	Substantial fraction	Tune packing, fluidity and adsorption
Cholesterol / neutral lipids	Smaller fraction	Modulate film properties
SP-A, SP-B, SP-C, SP-D	Protein fraction	Film organization, spreading and innate immunity

❖ *Exact percentages vary among sources and physiologic states; DPPC is the principal surface-active phospholipid.*

DPPC: molecular structure

- ❖ Dipalmitoylphosphatidylcholine (DPPC) is a glycerophospholipid and a major component of pulmonary surfactant.
- ❖ It contains a glycerol backbone, two palmitoyl (16:0) fatty-acyl chains, phosphate and choline.
- ❖ The molecule is amphipathic: a hydrophilic phosphocholine headgroup is attached to hydrophobic hydrocarbon tails.
- ❖ At the alveolar air–liquid interface, molecules orient with hydrophilic heads toward the aqueous phase and hydrophobic tails toward the air.
- ❖ Close packing of saturated DPPC molecules supports very low surface tension during compression.



(2) Lecithin (phosphatidylcholine)

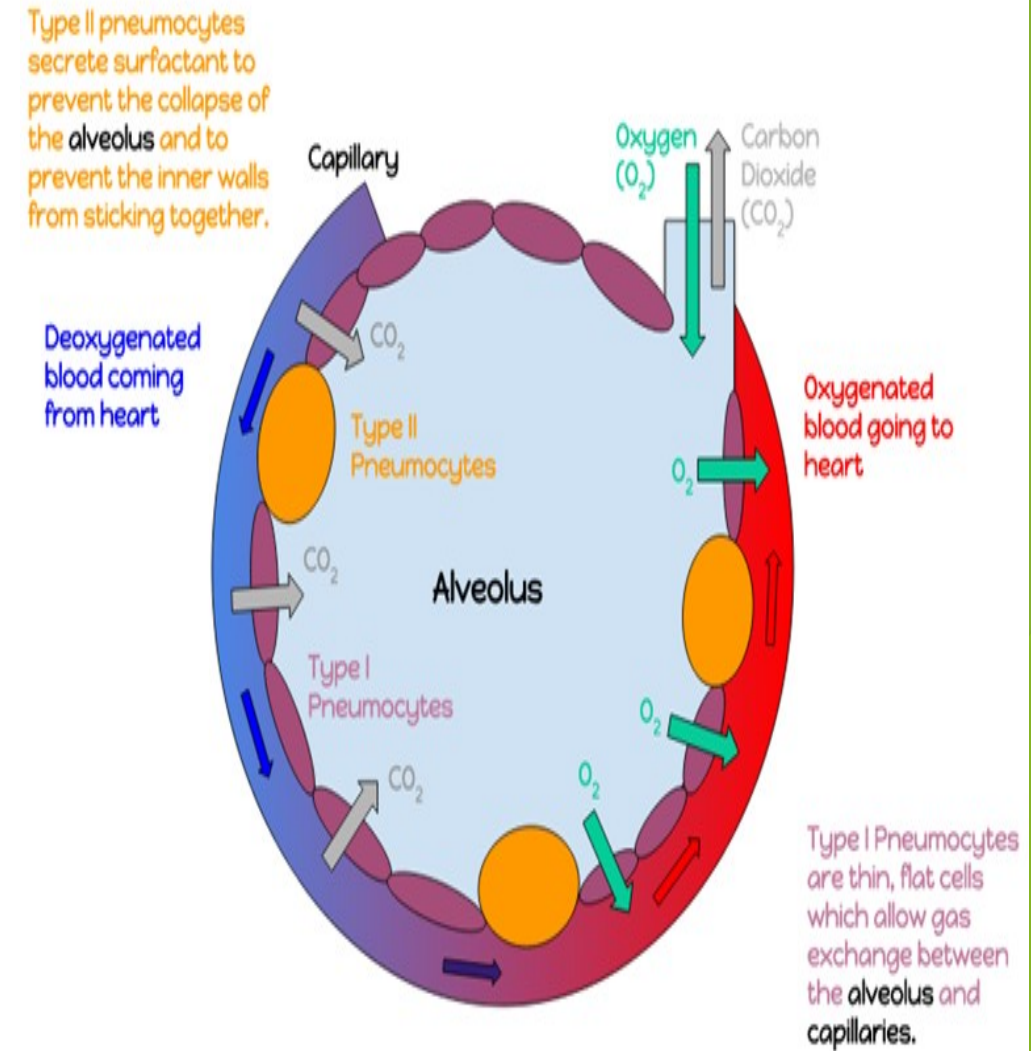
Surfactant proteins: SP-A, SP-B, SP-C and SP-D

Protein	General class	Main function
SP-A	Collectin	Innate immune recognition; surfactant organization and regulation
SP-B	Hydrophobic surfactant protein	Essential for film spreading and stability
SP-C	Hydrophobic surfactant protein	Promotes lipid film insertion, spreading and stability
SP-D	Collectin	Innate immune recognition and modulation of pulmonary inflammation

❖ *SP-B and SP-C are especially important for biophysical surfactant function; SP-A and SP-D have major host-defense roles.*

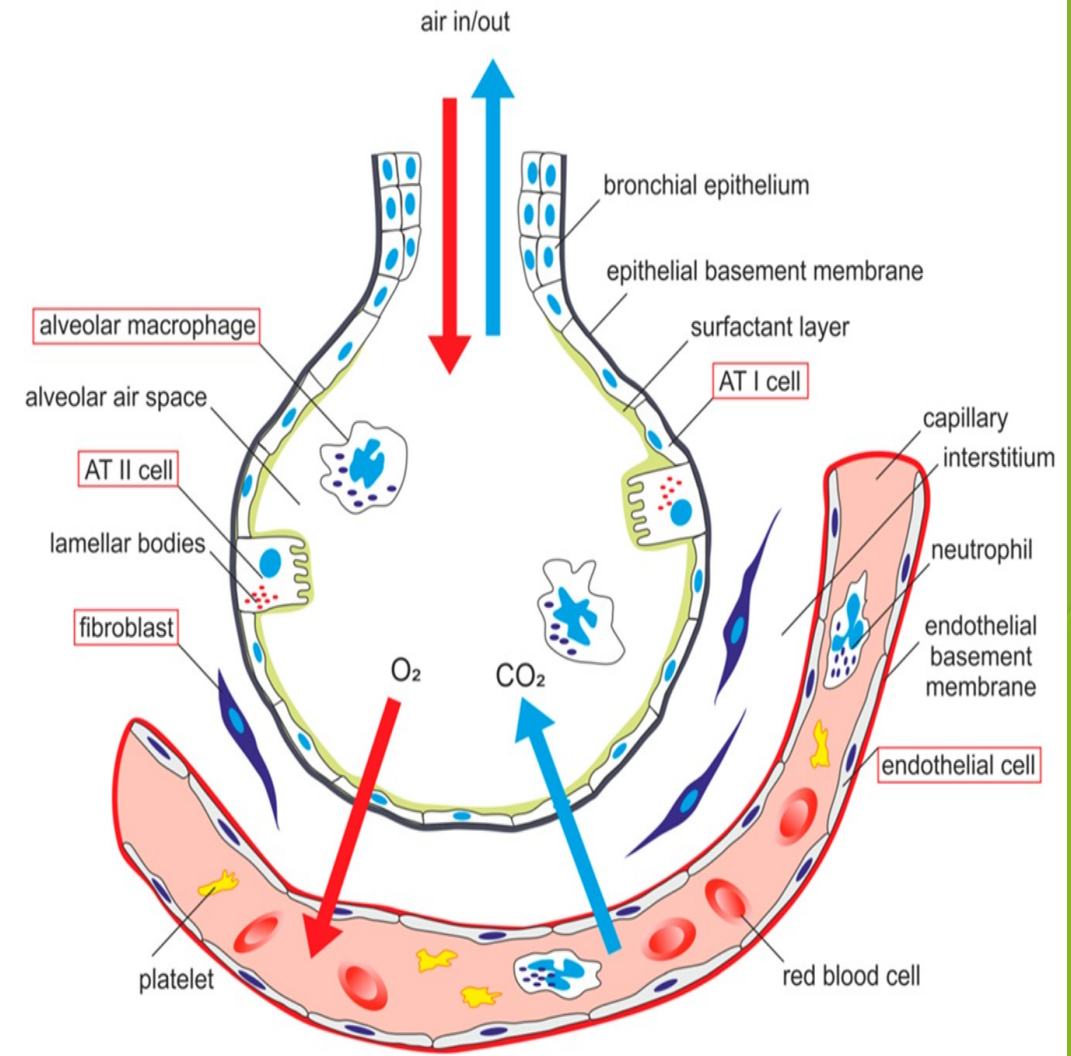
Surfactant film and surface tension

- Surfactant molecules adsorb at the alveolar air–liquid interface.
- During expiration, the surface area decreases and surfactant molecules become more densely packed.
- Lower surface tension reduces the collapsing force on alveoli.
- During inspiration, the film expands and surfactant components redistribute to maintain effective surface activity.
- Surfactant proteins and mixed lipids are required because pure DPPC alone does not have ideal adsorption and dynamic performance under physiologic conditions.



Alveolus and blood–air barrier

- ❖ Type I alveolar epithelial cells form most of the thin gas-exchange surface.
- ❖ Type II alveolar cells synthesize and secrete pulmonary surfactant.
- ❖ Pulmonary capillaries closely appose the alveolar epithelium.
- ❖ Gas crosses the alveolar epithelium, interstitial structures and capillary endothelium before entering plasma/RBCs.
- ❖ Diffusion is driven by partial-pressure gradients: O_2 moves toward blood; CO_2 moves toward alveolar air.



Type II pneumocytes and surfactant secretion

- ❖ Type II alveolar cells synthesize surfactant lipids and proteins.
- ❖ Surfactant components are packaged into specialized intracellular storage organelles called lamellar bodies.
- ❖ Lamellar bodies release surfactant into the alveolar lining fluid by exocytosis.
- ❖ Surfactant spreads over the air–liquid interface and is continuously recycled and degraded.
- ❖ Type II cells also serve as progenitor cells that can proliferate and contribute to alveolar epithelial repair after injury.

Fetal lung maturation and surfactant

- ❖ Surfactant synthesis begins during fetal development and increases substantially as gestation advances.
- ❖ Premature infants may have insufficient surfactant relative to the mechanical demands of the immature lung.
- ❖ Low surfactant increases surface tension, decreases lung compliance and predisposes to alveolar collapse.
- ❖ Clinically, antenatal corticosteroids are used in selected pregnancies at risk of preterm delivery to accelerate fetal lung maturation.
- ❖ Neonatal respiratory support is indicated.

❖ Important nuance

- The old lecithin:sphingomyelin ratio is historically important; modern fetal lung assessment may use additional biochemical markers and clinical context.

Surfactant deficiency and respiratory distress

Condition	Surfactant problem	Physiologic consequence
Neonatal RDS	Immature type II cells / insufficient surfactant	↑ surface tension, ↓ compliance, atelectasis
ARDS	Injury/inflammation can inactivate or disrupt surfactant	Loss of surface activity + impaired gas exchange
Pulmonary inflammation	Proteases, phospholipases and reactive oxygen species can damage components	Surfactant dysfunction
Atelectasis	Alveolar instability/collapse	Reduced ventilated surface area

Surfactant and innate immunity

- ❖ SP-A and SP-D are collectins that recognize carbohydrate-containing structures on microorganisms.
- ❖ They can aggregate pathogens and promote interactions with alveolar macrophages.
- ❖ Surfactant therefore links mechanical lung stability with host defense.
- ❖ SP-A also participates in regulation of surfactant homeostasis and inflammatory signaling.
- ❖ Damage to surfactant composition or function can therefore affect both gas exchange and pulmonary defense.

❖ Two major roles

- Biophysical: reduce surface tension. 2) Biological: support innate pulmonary defense.

Part IX — Integrated Biochemistry

Connecting protein structure, blood chemistry and respiratory physiology

Integrated pathway: O₂ from atmosphere to mitochondria

- ❖ Ventilation brings O₂ into alveoli.
- ❖ Surfactant stabilizes alveoli and reduces the work of breathing.
- ❖ O₂ diffuses across the blood–air barrier into plasma and RBCs.
- ❖ O₂ binds Fe²⁺ in Hb heme; oxygenation shifts Hb toward the R state.
- ❖ Blood transports O₂ to systemic tissues.
- ❖ Low pO₂, higher CO₂/H⁺, temperature and 2,3-BPG promote tissue unloading.
- ❖ O₂ enters cells and ultimately supports mitochondrial oxidative phosphorylation.



Integrated pathway: CO₂ from tissue to lungs

- Metabolism generates CO₂ and H⁺ in tissues.
- CO₂ enters RBCs and is converted rapidly to bicarbonate by carbonic anhydrase.
- Hemoglobin buffers H⁺ and carries a fraction of CO₂ as carbaminohemoglobin.
- Bicarbonate is the major circulating form of CO₂.
- In pulmonary capillaries, Hb oxygenation promotes H⁺ and CO₂ release (Haldane effect).
- CO₂ is regenerated from bicarbonate and diffuses into alveoli for exhalation.

❖ The two-way coupling

❖ O₂ transport and CO₂ transport are not separate systems: Hb allosteric couples them continuously.

High-yield structure–function relationships

Structural feature	Functional consequence
Monomeric Mb + one heme	Hyperbolic binding; O ₂ storage/facilitated diffusion
Hb tetramer $\alpha_2\beta_2$	Subunit communication and cooperativity
Fe ²⁺ in protoporphyrin IX	Reversible O ₂ binding
Proximal His–Fe bond	Couples heme movement to globin conformation
Distal His pocket	Supports controlled ligand binding
Central Hb cavity	2,3-BPG binding and T-state stabilization
Hb β_6 Glu→Val	HbS polymerization when deoxygenated
DPPC amphipathic structure	Surface-tension reduction at alveolar interface

Summary of Clinical significance Surfactant

- ❖ Insufficient surfactant in the alveoli is one of the causes that can contribute to atelectasis (collapse of part or all of the lung).
- ❖ The severe condition of acute respiratory distress syndrome (ARDS) is caused by a deficiency or dysfunction of surfactant
- ❖ Insufficient surfactant in the lungs of preterm infants causes infant respiratory distress syndrome (IRDS)
- ❖ The lecithin–sphingomyelin ratio is, a measure of fetal amniotic fluid to indicate lung maturity or immaturity
- ❖ A low ratio indicates a risk factor for IRDS.
- ❖ Lecithin and sphingomyelin are two of the phospholipid of pulmonary surfactant.

Summary of Clinical significance Surfactant Cont...

- ❖ Lung surfactant alterations have been described in a number of respiratory diseases.
- ❖ Surfactant deficiency in premature animals causes neonatal respiratory distress syndrome
- ❖ Mechanisms of Lung Surfactant Abnormalities
- ❖ Surfactant deficiency can be due to : an immaturity or injury to type II alveolar cells leading to insufficient surfactant synthesis, storage, or release
- ❖ Surfactant dysfunction or loss of surface activity can be caused by the degradation of surfactant components
- ❖ Degradation of surfactant components occurs via the
- ❖ action of proteases, phospholipases, or reactive oxygen species liberated during pulmonary inflammation

Key take-home messages

- ❖ Heme is the ligand-binding prosthetic group; globin creates the functional microenvironment.
- ❖ Myoglobin is a monomeric O₂-storage/facilitation protein with a hyperbolic curve.
- ❖ Hemoglobin is a tetrameric allosteric O₂ transporter with positive cooperativity and a sigmoidal curve.
- ❖ 2,3-BPG, H⁺, CO₂, temperature and chloride help regulate O₂ unloading.
- ❖ Bohr and Haldane effects explain the tight coupling between O₂ and CO₂ transport.
- ❖ HbS illustrates how a single amino-acid substitution can change protein–protein interactions and cell physiology.
- ❖ Pulmonary surfactant is a lipid–protein system that stabilizes alveoli and also contributes to innate defense.

Self-assessment questions

1. Why does myoglobin have a hyperbolic O₂ curve while hemoglobin has a sigmoidal curve?
2. What is the role of Fe²⁺, proximal histidine and distal histidine in the heme pocket?
3. Explain how O₂ binding at one Hb subunit influences the other three subunits.
4. Why does 2,3-BPG shift the Hb curve to the right?
5. Distinguish the Bohr effect from the Haldane effect.
6. Why can CO poisoning cause severe tissue hypoxia even when some Hb remains oxygenated?
7. How does $\beta 6$ Glu→Val produce the sickling phenotype?
8. Why is DPPC especially important in pulmonary surfactant?
9. What is the difference between surfactant deficiency in neonatal RDS and surfactant dysfunction in ARDS?
10. Explain how protein structure links molecular events to clinical manifestations.

Suggested references for further study

- Supplied lecture note: Structure Myoglobin & Hemoglobin; Oxygen & Carbon Dioxide Transport; Lung Surfactant.
- Harper's Illustrated Biochemistry — heme proteins, hemoglobin and oxygen transport.
- Lehninger Principles of Biochemistry — protein structure, ligand binding and allostery.
- Devlin's Textbook of Biochemistry — heme proteins and clinical biochemistry.
- Standard medical biochemistry texts for HbA1c and hemoglobinopathies.
- Standard physiology texts for gas exchange, ventilation and oxygen transport.

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THANK YOU

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